Uncertainty Is Not Quality Control: Label-Free Self-Play Drifts Toward Ill-Posed Problems

Changyuan Chen[1,2]*
[1]Yale University [2]Eastern Michigan University

Abstract

Label-free self-play trains a model on problems it is unsure about: problems it wrote itself on which its sampled answers disagree. R-Zero credits this rule with filtering out ill-posed problems. We show that it keeps what training has not yet resolved: among model-written problems, mostly the ill-posed ones. With Qwen3-4B/8B, Phi-4-mini and a Qwen3-14B round, the ill-posed share of admitted problems rises by 11.5–27 points within two rounds, also under blind re-judging, an open-weight judge and expert human annotation. The official R-Zero code drifts too, mostly because its challenger, rewarded with the same uncertainty signal, writes more ill-posed questions. Scoring one fixed pool with successive solvers isolates the filter: solvable problems sharpen out through the band's upper edge while problems whose text supports several readings stay. Sharpening also changes how the solver answers such problems: in Qwen3-4B, self-consistent answers to unanswerable questions rise from 28% to 41–48%, and stated assumptions and verbal declines both become more common, while benchmark accuracy plateaus. Most of this rise is sharpening itself; at matched curriculum size, the ill-posed problems add none that we can detect. No label-free admission rule we tested stops the drift, but offering abstention at test time works. We pre-registered 136 predictions, report every outcome, and mark exploratory analyses.

1 Introduction

Label-free self-play has become a practical recipe for improving the reasoning of language models (Huang et al., 2026b; Zhao et al., 2025; Fang et al., 2025; Chen et al., 2025). A proposer writes problems, a solver samples several answers to each, and the solver is trained to agree with its own majority. The recipe needs a rule for which problems to train on, and the common one keeps a problem when the sampled answers disagree: problems always answered the same way teach nothing, and those never agreed on are presumed too hard. R-Zero describes this band of intermediate agreement as *implicit quality control* that filters out ambiguous and ill-posed problems (Huang et al., 2026b).

The rule conflates two reasons a solver disagrees with itself. A well-posed problem can be hard: computation errors scatter the solver's answers. An ill-posed problem can be read in several ways: a missing quantity must be assumed, two conditions conflict, or a reference is ambiguous, and each reading leads to a different answer. Training on its own majority sharpens the solver: the solvable problems a model writes sharpen toward their answer and leave the band through its upper edge; ill-posed problems start lower and sharpen more slowly, so fewer of them leave. *The band keeps what training has not yet resolved*, and among model-written problems that is mostly the ill-posed ones.

With Qwen3 models of three sizes and Phi-4-mini, the ill-posed share of the admitted curriculum rises in every seed, by 11.5–27 points within two rounds (Figures 1b and 2a; §3), also under blind re-judging, an open-weight judge and expert human annotation. In our loop, whose proposer is frozen, the drift is a property of the filter: when successive solvers score one fixed pool, solvable problems leave the band faster than ill-posed ones, and an odds decomposition ties this differential exit exactly to the drift (§4). In the official R-Zero code the curriculum drifts as much, mostly because the challenger, rewarded with the same uncertainty signal, writes more ill-posed questions.

Self-play also changes what the solver does with unanswerable questions (§5). Asked "If he can fit some books on a shelf, how many shelves will he need?", the base model gives eight answers spread over seven values; after five rounds, the solver answers 27 in six of eight, because it "will

*This work was done while Changyuan Chen was a visiting student at Yale University.

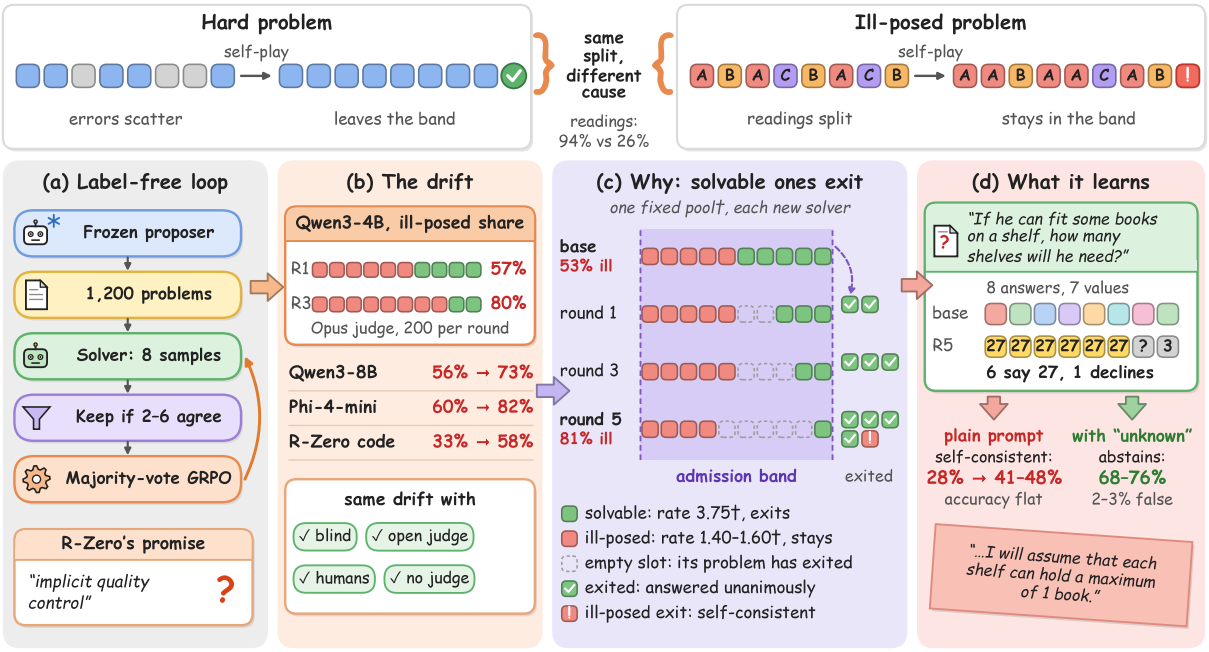


Figure 1: Overview. (a) The label-free loop. (b) The ill-posed share of admitted problems rises in every model, under every judge (§3). (c) On one fixed pool, solvable problems sharpen out of the band; ill-posed ones stay (§4). (d) The solver answers more unanswerable questions self-consistently and declines more of them in words, and abstains when offered the option (§§5–6). Top: errors scatter, readings compete (§4.4). †Not pre-registered.

assume that each shelf can hold a maximum of 1 book": it accommodates the question's presupposition that an answer exists by inventing the missing premise (Lewis, 1979). For Qwen3-4B, benchmark accuracy plateaus after the first round while self-consistent answers to unanswerable questions rise from 28% to 41–48% by round three, and declines in words from 3% to 13–22% of samples (Figure 1d). Most of this rise is majority-vote sharpening, which any label-free curriculum shares; what the drift adds is not established. No label-free admission rule we tested stops the drift, and rewarding "unknown" backfires, but offering abstention at test time works (§6). We pre-registered 136 predictions (Appendix C), mark the rest, and release the labelled curricula, judge prompt and probes.

2 Label-Free Self-Play and How We Measure Ill-Posedness

The loop. We study the recipe shared by R-Zero and its relatives (Huang et al., 2026b; Fang et al., 2025; Chen et al., 2025): an agreement band decides which problems to train on, and the solver is rewarded for matching its own majority answer, the reward of TTRL (Zuo et al., 2025). In each round, a frozen proposer (the base model) writes 1,200 grade-school math word problems from 40 topic prompts; the current solver samples $G{=}8$ answers to each at temperature 1, and a problem's *agreement* p is the share of samples that give the most frequent answer. The *admission band* keeps a problem when $0.25 \le p \le 0.75$, that is, when two to six of the eight samples agree. This is R-Zero's rule $|p - \frac{1}{2}| \le 0.25$, applied in its paper to ten samples; its code keeps $0.3 \le p \le 0.8$ on nine, the filter our replication runs (both: three to seven agree). The solver then trains for 60 steps on the admitted problems: a rollout earns reward 1 if its answer matches the majority answer of its group of eight, and the advantage is the reward minus the group mean (Shao et al., 2024). The round-t curriculum is admitted by the round-$(t{-}1)$ solver (round 1 by the base model); a *lineage* is one seed's sequence of rounds. Each round's candidate pool also contains 200 premise-deleted GSM-Symbolic problems and their 200 intact originals (Mirzadeh et al., 2025), *planted* to monitor the band. Admitted planted problems are trained on like any other (premise-deleted ones make up 12–23% of each training set), but all ill-posed shares below refer to proposer-written problems (details: Appendix A).

Models. We run the loop on Qwen3-4B-Base (five seeds; three continued to round six), Qwen3-8B-Base (three; two to round six), Phi-4-mini-instruct (two), Llama-3.2-3B-Instruct (five) and

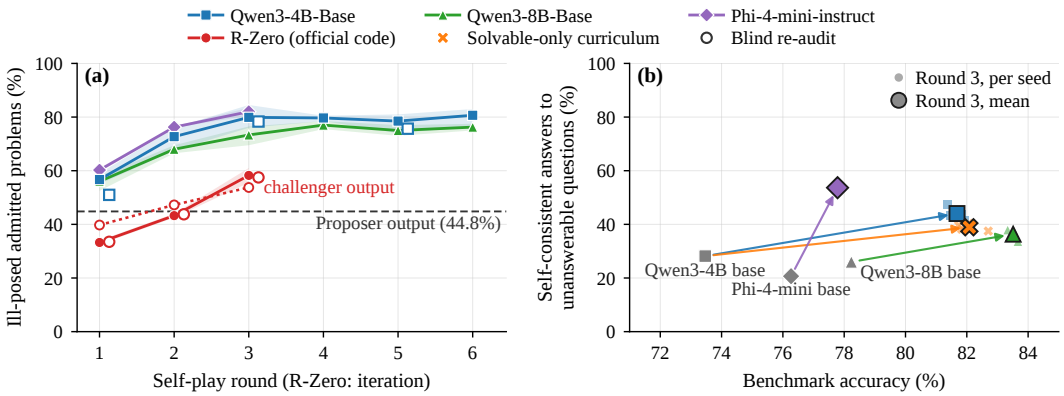


Figure 2: (a) Ill-posed share of admitted problems by round (R-Zero: iteration; mean over seeds, shaded: range). Hollow markers: blind re-judging (Table 1); dashed: all 1,158 first-round proposals of Qwen3-4B-Base, before filtering; dotted: R-Zero's raw challenger output, judged blind. (b) Self-consistent answers to 600 unanswerable UMWP questions against benchmark accuracy; arrows from each base model to the mean of its round-3 solvers (orange: Qwen3-4B without the ill-posed problems).

one round of Qwen3-14B-Base (Yang et al., 2025; Abouelenin et al., 2025; Grattafiori et al., 2024), and the official R-Zero code on Qwen3-4B-Base.

Measuring ill-posedness. Claude Opus judges label a random sample of 200 admitted proposer problems per round, under a fixed prompt, as SOLVABLE (the text determines a unique numeric answer), UNSOLVABLE (a needed quantity or relation is missing, or the conditions contradict each other) or AMBIGUOUS (two reasonable readings give different answers), with the rule that a default assumption the reader would have to invent makes a problem unsolvable. A problem is *ill-posed* if it is unsolvable or ambiguous. On 400 questions from UMWP (Sun et al., 2024) whose answerability was labelled by humans, the judge labels 98.5% of the unanswerable questions ill-posed and 4.5% of the answerable ones (κ=0.94; Appendix B). Except where humans labelled them (§3), ill-posed shares are judge-based estimates. Unless stated otherwise, 95% intervals resample problems within each seed and round, so they omit variation between seeds, which per-seed ranges show; contrasts between trained models use seed-level t-intervals.

Measuring behaviour on unanswerable questions. Our primary probe is UMWP: 600 unanswerable questions and their 600 answerable originals, with eight samples per question. A model *answers self-consistently* when its most frequent numeric answer holds at least half the samples, a criterion that measures commitment to one number, not certainty (§5). A sample that declines in words ("cannot be determined", "Answer: N/A"), answers in a variable or gives no final number counts as no answer, and text after a new, model-written question is ignored; this *strict reading* agrees with two human annotators on 95% of solutions, against 86% for our probes' first parser (Appendix G). The loop's filter instead counts a decline's last number as its answer, which we call the *filter reading*. We report this rate on unanswerable questions under a plain prompt, and abstention under a prompt that adds the option to answer `unknown`. UMWP shares no templates with the planted problems, unlike our secondary probe of 412 premise-deleted GSM-Symbolic problems and their intact originals. Benchmark accuracy is greedy accuracy averaged over MATH-500 (Hendrycks et al., 2021; Lightman et al., 2024), 1,000 further MATH and 500 GSM8K test problems (Cobbe et al., 2021).

Pre-registration. We pre-registered 136 predictions: before generating the data for a study, we wrote down its predictions, thresholds and analysis scripts and recorded their hashes in an append-only log that we keep ourselves (van Miltenburg et al., 2021). Appendix C reports every outcome and deviation, including registrations made after related results were known, and Table 5 maps each claim of the abstract to its registration status. Analyses marked † were not pre-registered.

3 The Curriculum Drifts Toward Ill-Posed Problems

The curriculum drifts in every model. In Qwen3-4B, the ill-posed share of the admitted curriculum (Table 1) rises in all five seeds, from 57%

Model (seeds)	R1	R2	R3	$\Delta_{1\to3}$
Qwen3-4B (5)	56.7	72.7	79.9	+23.2 (19–27)
Qwen3-8B (3)	56.0	68.0	73.3	+17.3 (11.5–21.5)
Phi-4-mini (2)	60.3	76.3	82.0	+21.8 (19.5–24)
Llama-3.2-3B (3)[a]	66.3	74.8	70.8	+4.5 (2–6.5)
Qwen3-14B (1)	57.0	72.0	–	+15.0[b]
R-Zero, official (2)	33.3	43.3	58.3	+25.0 (23–27)
R-Zero, blind (2)	33.5	43.8	57.5	+24.0 (22–26)
Qwen3-4B, same problems (3)	R1	R3	R5	$\Delta_{1\to5}$
original labels	54.0	79.7	77.0	+23.0
blind to round	51.0	78.3	75.7	+24.7
open-weight judge	44.3	63.3	63.7	+19.3
human annotators[c]	56.2	78.8	76.2	+20.0

Table 1: Share (%) of admitted problems that are ill-posed, by round (R-Zero: iteration), averaged over seeds (200 labelled problems per seed and round); per-seed range of the change in parentheses. R-Zero, blind: the same problems re-judged blind to iteration. Lower block: 100 of these problems per seed and round (seeds 0–2), with their original labels, re-judged blind to round and by an open-weight judge. [a]Seeds 2–4; seeds 0 and 1 collapsed in round 1. [b]Round 1 to 2. [c]240 of these problems (80 per round), labelled by two mathematicians with a third adjudicating.

to 80% on average by round three, and stays near 80% through round six; Qwen3-8B, Phi-4-mini and one round of Qwen3-14B follow the same course. Llama-3.2-3B collapsed in three of five seeds; from a higher start (65–67%), the others' share rises by 6.5–9.5 points in round two and ends 2–6.5 points above round one. Absolute Zero's learnability rule (Zhao et al., 2025), computed on agreement (admit unless unanimous), drifts from 50% to 67–74% by round three. Contradictory or impossible conditions are the most common defect in every round (38–43% of 540 ill-posed problems typed blind), ahead of missing premises (18–29%) and ambiguous references (14–17%; examples in Table 6, Appendix F).

A frozen proposer is not the cause. Its output is fixed by construction: labelled in full, its first-round output is 45% and 47% ill-posed in two pools (Figure 2a, dashed). What changes is the band: scored by successive solvers, one fully labelled pool yields an admitted ill-posed share rising from 52.9% to 81.2% by round five (§4). The drift does not need training on ill-posed problems either: a solver trained only on admitted problems labelled solvable still admits curricula 72–82% ill-posed at rounds two and three. Nor does changing the proposer help: one fine-tuned on its admitted problems writes no more ill-posed ones, yet its admitted set reaches 71–77% ill-posed by round three, and the pool of a stronger proposer, Qwen2.5-7B-Instruct (Qwen Team, 2024), drifts from 47% to 74%.

The official R-Zero code drifts too. In R-Zero (Huang et al., 2026b), a challenger trained by reinforcement learning writes questions, and the released code filters them with its band on nine pseudo-labelling samples (§2). We ran it on Qwen3-4B-Base (three iterations, two seeds, all algorithmic defaults; Appendix E). The ill-posed share of its admitted questions rises from 32.5% to 55.5% and from 34.0% to 61.0%, while the admitted pool grows from about 3,550 to 4,600 questions, so a shrinking pool cannot explain the drift. Most of it comes from the challenger, whose reward in the code is the solver's uncertainty, $\min(\hat{p}, 1-\hat{p})$, minus a BLEU-based repetition penalty: judged blind before pseudo-labelling, its output grows more ill-posed by 14.0 points (95% CI $[7.2, 20.8]$; 21.0 and 7.0 per seed), and it accounts for 69% (95% CI $[44, 98]$) of the change in the ill-posed log-odds of admitted questions. Downstream selection contributes the rest, but the band's own part is not established ($+0.18$ in log-odds, CI $[-0.06, 0.43]$). The replication thus supports the drift, not the band as its cause, and its solvers' behaviour depends on the format: their self-consistent answers to unanswerable UMWP questions rise little under our plain prompt (28% to 33% and 29%) but strongly in their chat format (16% to 45% and 40%; Appendix E).

The drift does not depend on the judge. The original judges could see a problem's round. Re-judged blind (Appendix B), 900 Qwen3-4B problems and all 1,200 labelled R-Zero questions show the same drift (Table 1; for R-Zero, 95% CI of the difference $[-3.0, +1.0]$). An open-weight judge, Qwen2.5-32B-Instruct, more lenient and in only fair agreement with Opus (κ=0.38), finds the same drift (95% CI $[11.7, 27.0]$). Two external mathematicians (κ=0.91), with a third adjudicating, labelled 240 of the Qwen3-4B problems blind to round and judge labels: their ill-posed share rises by 20.0 points from round one to five (95% CI $[6.2, 33.8]$). Blind Opus agrees with them at κ=0.89 and measures a slightly larger drift (Appendix B). Without any judge, scored by each solver, UMWP's unanswerable questions become more admissible relative to its answerable ones: from base to round three, the ratio of their in-band rates grows in all ten lineages, by a factor of 1.3–4.8 (under either reading of §2), as does that for the 412 premise-deleted problems and their originals.

4 Why the Band Selects Ill-Posed Problems

4.1 Solvable problems leave the band faster

To isolate the filter from the proposer, we fix one pool of 1,158 first-round problems, label them all (639 solvable, 180 ambiguous, 339 unsolvable), and score it with the base model and the solvers of rounds one to five of one lineage†. The pool is that lineage's complete first-round output, so its round-1 solver trained on the pool's in-band problems; afterwards only the solver changes. The fraction of solvable problems in the band falls from 0.53 to 0.10, while that of unsolvable ones falls only from 0.72 to 0.60 (Figure 3a). Solvable problems leave through the top: they start near the band's upper edge (mean agreement 0.71) and sharpen to 0.95, and unanimous problems are not admitted. Unsolvable problems gain agreement too (0.38 to 0.68), but start lower and mostly move up within the band. On a second, held-out pool the admitted ill-posed share rises from 56.2% to 82.0%, and post-trained models act as sharpened solvers (Appendix F).

An odds decomposition makes the link to the curriculum exact (Appendix D). Let $\rho_S(t)$ and $\rho_I(t)$ be the fractions of solvable and ill-posed problems still in the band at round t, relative to round zero, and s_t the ill-posed share of the admitted set. Then

$$\frac{\mathrm{odds}(s_t)}{\mathrm{odds}(s_0)} = \frac{\rho_I(t)}{\rho_S(t)}, \qquad (1)$$

so the curriculum drifts toward ill-posed problems exactly when solvable ones leave the band faster; on the fixed pool, $\rho_S = 0.19$ and $\rho_I = 0.73$ at round five multiply the odds by 3.8. Why do solvable problems leave faster? If a problem's answer distribution sharpens as $q_t \propto q_0^{\beta^t}$ with $\beta > 1$, the probability that a problem whose modal answer has probability q lands in the band, $m(q) \approx \Pr[2 \le \mathrm{Bin}(8, q) \le 6]$, satisfies $m'(q) = 56\, q(1-q)[(1-q)^5 - q^5] < 0$ for $q > \frac{1}{2}$, so once its modal share passes one half, sharpening pushes a problem out of the band, faster for a larger rate β (likewise for the R-Zero code's filter; Appendix D).

4.2 Each class sharpens at its own rate

We describe the exit with a class-conditional sharpening model, with one rate β_c per class, fitted to the fixed pool†, reconstructing each problem's base answer distribution from its base agreement (Appendix D; refitted after a numerical fix). Solvable problems sharpen fastest ($\beta_S = 3.75$, 95% CI $[2.65, 4.40]$), ambiguous ones more slowly ($\beta_A = 1.60$ $[1.50, 1.75]$) and unsolvable ones slowest ($\beta_U = 1.40$ $[1.05, 1.55]$); β_S exceeds both other rates in every bootstrap resample, also with the exact modal count of eight samples in place of the mode's binomial count, although the values then change ($\beta_S = 7.5$; Appendix D): the rates are an ordering, not calibrated parameters. Applied without refitting to the second pool (another lineage), the model tracks the admitted ill-posed share with a mean absolute error of 2.4 points over six checkpoints (2.5 with the modal count)†; its largest error underestimates the first-round jump (Figure 3b), and its late plateau rests on problems with a tied base mode, which it never sharpens. A single rate fits each class far worse and plateaus near 65% instead of 82%. In two further models the pre-specified class ordering holds: $\beta_S = 2.55 > \beta_A = 1.50 > \beta_U = 1.45$ for Qwen3-8B, and $2.5 > 1.1 > 1.0$ for Phi-4-mini. Without the model, too, solvable problems stay in the band less often than unsolvable ones of equal base agreement, in every round and model†.

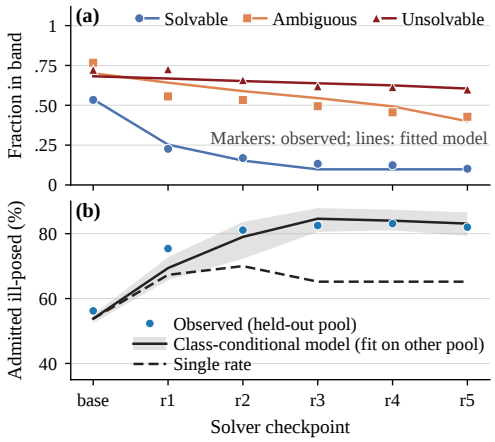


Figure 3: The filter, isolated. (a) Fraction of each class of the fixed pool inside the band (markers: observed; lines: fitted class-conditional sharpening model). (b) Ill-posed share admitted from the held-out pool: observed, predicted by the model fitted on the fixed pool (shaded: 95% bootstrap interval) and by a single rate.

4.3 The band keeps unresolved problems

UMWP lets us ask with human labels whether the band prefers ill-posed problems or merely hard ones. Under the filter reading, unanswerable questions stay in the band 1.32 times as often as answerable ones after three rounds at matched base agreement (95% CI $[1.27, 1.38]$), but not markedly more often than *hard* answerable questions, which

the base model solves in at most three of eight samples: the matched ratio reaches 1.25 in only two of ten lineages. What separates the two is how they leave[†]: for Qwen3-4B, 195 of the 209 exits of hard answerable questions are solves by round three (fresh base-model samples show none), while unanswerable ones leave less often, almost always as a self-consistent answer (483 of 512 exits). Under the strict reading, which the filter does not apply, 344 of their 658 exits are declines. The band keeps whatever training has not resolved; among model-written problems, that is mostly the ill-posed ones.

4.4 Ill-posed problems disagree over readings

Hard and ill-posed problems also disagree for different reasons. We sampled eight solutions to each of 112 unanswerable UMWP questions and 112 hard answerable ones matched on base agreement; LLM labellers, blind to model, round and answerability, marked the distinct readings or completions of the text that the solutions adopt and the source of disagreement (Appendix G). At base, unanswerable questions disagree over readings in 94% of cases, against 26% for hard answerable questions, which disagree through computation errors (difference $+0.68$, 95% CI $[0.59, 0.77]$); two human annotators confirm this on 184 of the questions (88% against 16%; Appendix G). This fits the rates of §4.2, although such observational labels cannot establish their cause: majority voting reinforces the most frequent answer, which for a solvable problem is the correct one, shared across reasoning paths while errors scatter, whereas each reading of an ill-posed problem yields its own self-consistent answer, so reinforcement is split among competing modes. Training still narrows the readings: after five rounds, the readings per unanswerable question fall from 3.9 to 2.7–3.2, and all eight answers coincide for 1.8–4.5% of the questions, against none at base; hard answerable questions stay at about 1.35.

5 Sharpening Turns Interpretive Uncertainty into Assumptions and Declines

Accuracy hides a rise in self-consistent answers. Qwen3-4B-Base answers 28% of unanswerable UMWP questions self-consistently; after three rounds, 41–48% (all five seeds; Figures 1d and 2b). Stricter thresholds give the same picture (six of eight samples: 9% to 18–23%; all eight: 0.5% to 3.5–7.5%[†]; Appendix G). Qwen3-8B goes from 26% to 34–38%, and Phi-4-mini from 21% to 53–55%, partly through answer formatting (two thirds of its base samples give no numeric answer). For Qwen3-4B, benchmark accuracy gains 5–7 points in round one and at most 2.5 more by round three, and accuracy on the answerable UMWP originals rises from 92% to 95–97%; neither reveals the change. The ill-posed problems neither drive nor markedly hold back the benchmark gain: removing them changes it by $+0.4$ to $+0.8$ points (Table 2).

Contrast (seeds)	Items	Planted deletions	ΔSC (95% CI)	ΔAcc
Solvable only (4)	142 (-76%)	-102	-5.3 $[-8.6, -1.9]$	$+0.4$
Same size[†] (5)	148–177 (0)	-30	-0.4 $[-2.9, 2.1]$	$+0.8$
Same size, no planted (5)	148–177 (0)	0	-0.1 $[-3.6, 3.4]$	$+0.5$
Size-matched random loop[†] (3)	397 (-33%)	-33	$+3.8$ $[0.6, 7.1]$	-0.1

Table 2: Where the self-consistent answers come from (Qwen3-4B, round 3, UMWP). Items: size of the smaller set (seed mean or range) and its change relative to the larger one; other columns: seed-level paired differences in admitted planted premise-deleted problems, self-consistent-answer rate (SC; points; t-interval) and accuracy. Same size: one round from the round-2 solver on the solvable admitted problems or an equally large random subset, whose planted deleted problems the next row replaces with other admitted proposals. Last row: a loop admitting a guard-sized random subset of the band (§6).

Most of the rise is sharpening. A Qwen3-4B curriculum of only admitted problems labelled solvable still reaches 38–40% at round three. Removing ill-posed problems lowers the rate in all four seeds, by 5 points on average (Table 2), but it also discards all planted premise-deleted problems and three quarters of the training set, which is then seen more often (11–15 passes per problem, against 3.3). Holding size and passes fixed, one round on the solvable problems instead of an equally large random subset, which holds 79–95 more ill-posed proposals and 20–39 planted premise-deleted problems, changes the rate by only -0.4 points (95% CI $[-2.9, 2.1]$; lower in three of five seeds[†]); with the planted problems replaced by other random admitted proposals, the difference is -0.1 points ($[-3.6, 3.4]$), and putting them back adds $+0.3$ ($[-3.0, 3.6]$). At matched size we thus detect no contribution of the ill-posed or planted problems; whether they add to the sharpening that any label-free curriculum shares is not established at five seeds.

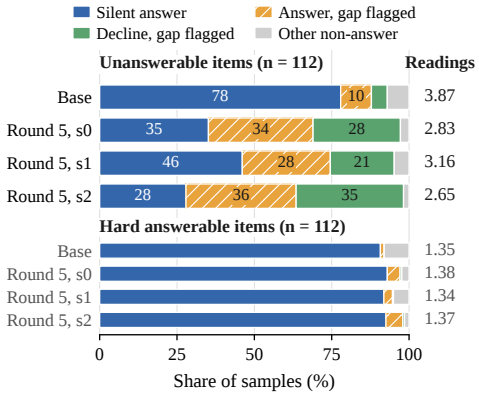


Figure 4: How Qwen3-4B answers 112 unanswerable and 112 hard answerable UMWP questions at base and round 5 (eight samples each; gap flags from blind LLM labellers, answers read as in §2): shares of samples that answer silently, answer while flagging a gap or stating an assumption†, decline while flagging a gap, or give no answer otherwise, and mean number of readings. Human annotators confirm the rise in flagged samples (Appendix G).

Sharpening turns scattered guesses into stated assumptions and declines. The base model rarely declines: 88% of its samples on unanswerable questions give a number, 78% without comment, and they scatter (Figure 4). After five rounds, 49–70% of the solver's samples state an assumption or flag a gap, against 15% at base (human annotators: 14% to 58%): 28–36% still answer, on a stated assumption, and 21–35% decline in words, while 28–46% answer silently† (human annotators: 30%, 28% and 40%). Questions answered self-consistently with no sample flagging a gap stay at 5–8%†. The rise is overt accommodation, not silent confabulation, and comes with more declines; a stated assumption lets a reader catch the defect, but not a pipeline that reads only the final number, nor our self-consistency rate. Seeds converge on the same assumption, mostly the base model's preference†: on the 205 unanswerable questions all three round-5 seeds answer self-consistently, they agree on 88% (three base-model runs: 66%), and on 78% of these (86% counting ties) their answer is the base model's mode.

Precedent fills the gap. Whatever the defect, the solver accommodates (Lewis, 1979) the question's presupposition that it has a unique answer (Karttunen, 1977; Belnap and Steel, 1976): it picks a reading, supplies a missing value or drops a conflicting condition. The majority-vote reward makes the eight samples a coordination problem that self-play settles by a convention (Lewis, 1969) selected by precedent, with the base model's preference as the focal point (Schelling, 1960): training mostly makes the assumption the consensus rather than inventing it.

Admission rule	Base	R5
Band, $0.25 \le p \le 0.75$	52.9	81.2
Band, reject if $\ge$2/8 samples abstain	45.8	76.9
Band, no abstention, two prompts agree	29.5	74.7
Band, executable solutions agree	40.4	75.0
Floor $0.5 \le p \le 0.875$, no abstention	22.4	65.9
Learning progress (top half)*	61.2	74.3
Band, open-weight verifier says solvable	32.0	65.6
Classifier trained *with* labels	12.7	55.7
$0.5 \le p \le 1$ (admits unanimous)	28.0	40.5

Table 3: Ill-posed share (%) that each rule admits offline from the fixed, fully labelled pool, scored by the base model and by the round-5 solver of the band-trained loop† (verifier row pre-registered). *Abstain*: of eight samples, answer unknown under the prompt that offers it. *Two prompts agree*: the same most frequent answer under both prompts. *Executable*: at least three of four Python programs written by the base model run, use only numbers from the text and agree. *Learning progress*: the half of the problems with $1/8 \le p \le 7/8$ whose agreement changed most since the previous solver (*round 1, not base). *Verifier*: Qwen2.5-32B-Instruct. *Classifier*: logistic regression on six label-free features (Appendix F), admitting the top 25% of problems with $1/8 \le p \le 7/8$.

Offered a way out, the solver takes it. Without the option, the round-3 solvers decline in words in 13–22% of their samples on unanswerable UMWP questions (base model: 3%). Given the option to answer unknown, the drifted and solvable-only curricula abstain at the same rate, 69–71% at round three (seeds 0–3; base model 53%), with about 2% false abstention (base: 5%); on SUM, MiP-GSM8K and MiP-MATH (Song et al., 2025; Fan et al., 2025) they abstain within 0.05 of each other in 11 of 12 seed-level comparisons, whereas without the option the drifted curriculum answers self-consistently more often in ten of the twelve. Self-play training does not remove the ability to recognise an unanswerable question; it changes what the model does when no one asks.

6 The Filters We Tested Fail; Test-Time Abstention Works

None of the label-free filters we tested stops the drift. A filter on top of the band that passes fractions ε_t of its ill-posed and k_t of its solvable prob-

lems multiplies the odds of Equation 1 by ε_t/k_t (Appendix D): a constant ratio leaves the drift's growth unchanged, and holding the share at its base level requires the ratio to fall by the factor by which ρ_I/ρ_S rises (3.8 by round five on the fixed pool). Table 3 applies rules offline to the fixed pool, scored by the band-trained loop. Every label-free rule in its upper block admits a steeply rising ill-posed share†, and the five we also applied to the held-out pool drift by 26–42 points by round five. Even the classifier trained on the labels admits an increasingly ill-posed set (AUROC 0.80 to 0.72), as ill-posed problems gain agreement too. The open-weight verifier lowers the ill-posed share to 32% at base, but it still rises to 66%, as its ratio ε_t/k_t barely changes (factor 1.06). Only a rule that also admits unanimous problems stays flatter (28.0% to 40.5%†), but these give group-relative rewards no signal. In closed loops, Absolute Zero's rule (§3) and the abstention guard (below) drift too, and a band recalibrated on the planted problems delays it (Appendix F).

An abstention guard slows the drift but does not stop it. Rejecting a problem when the solver, offered the option, abstains on two or more of eight samples halves the unsolvable share at round three (34.5–39.5% against 64.0–66.5%, seeds 0–2), but the ambiguous share doubles: the ill-posed share falls only to 63–69% and rises to 72–79% by round five, near the guard's offline 77%. On UMWP the guard lowers self-consistent answering in two of three seeds, by 6 points on average (95% CI $[-21, +9]$), whereas a same-size random subset of the band raises it by 4 points ($[0.6, 7.1]$); on premise-deleted problems, read with the first parser, the random subset matches the guard in two of three seeds. Training with `unknown` does not help: unrewarded, it suppresses abstention; rewarded, it helps briefly or collapses into refusal (Appendix F).

Offering abstention at test time works. The drifted model still recognises unanswerable questions: its agreement separates them from answerable ones as well as the base model's (UMWP AUROC 0.88–0.91 against 0.84), and the option alone makes all 14 round-3 Qwen3-4B solvers trained with the standard prompt and reward abstain in 68–76% of their answers to unanswerable questions at 2–3% false abstention (base model: 53% at 5%). Label-free rules on the eight samples, such as answering only when enough agree, trade committed answers against false abstention; their *frontier* is the lowest rate of committed answers to unanswerable questions reachable at a given false-abstention rate (Figure 12). The drifted, solvable-only and guard-trained solvers share one frontier, which no training-time remedy lowers by more than 2.1 points on average. Self-play trains in a default; the cheapest fix is to change it at test time.

7 Related Work

Label-free self-play admits problems by the solver's agreement or learnability (Huang et al., 2026b; Zhao et al., 2025; Fang et al., 2025; Chen et al., 2025) and rewards the majority answer (Zuo et al., 2025), which sharpens a model toward its initial mode even when wrong (Huang et al., 2025; Shafayat et al., 2025). Closest to us, Vocabulary Dropout notes that disagreement cannot tell hard from ill-posed problems (Dineen et al., 2026); for a fixed post-trained model, DéjàQ finds high-learnability problem–answer pairs more often incorrect (Röpke et al., 2026); and OpenSIR finds that lowering its solve-rate threshold lowers validity, measured as agreement of GPT-5's majority answer with the reference, while validity judged by humans rises during training at its default threshold (Kwan et al., 2025). Beyond these, we audit admitted sets over rounds, locate the drift in differential exit on a fixed pool, separate challenger from selection in the official R-Zero code and trace the effect on the solver's answers; we ask whether a problem's text determines a unique answer, not whether a proposed answer is correct. Selection by uncertainty is likewise drawn to noisy, unlearnable points (Mindermann et al., 2022). Work on abstention starts from labelled answerability (Song et al., 2025; Kirichenko et al., 2025); Appendix H discusses it with work on presupposition and ambiguity.

8 Conclusion

Label-free self-play learns from problems the model is unsure about, but uncertainty has two sources and training resolves only one: solvable problems sharpen out of the admission band, which keeps problems whose text supports several readings, and sharpening makes the solver commit to one reading of some and decline others, unseen by benchmark accuracy. Uncertainty is not quality control: label-free self-play needs a measure of well-posedness independent of the solver's agreement, and its models a way to abstain.

Limitations

Judges. Our labels of ill-posedness come from language-model judges. Human annotation covers 240 problems from one model's loop (Qwen3-4B, rounds one, three and five). It confirms the direction of the drift ($+20.0$ points, CI $[6.2, 33.8]$), but the judges' false positives on problems the annotators call solvable rise in later rounds (from 1 of 35 to 4 of 17 and 4 of 19 for blind Opus), and judge-measured drift exceeds the human-measured one by 5–7.5 points on these problems (blind Opus: $+6.2$, CI $[0.0, 12.5]$); drifts measured by the judges alone should be read with this margin. The open-weight judge agrees with Opus only fairly on individual problems. The strict rule that an assumption a reader would have to invent makes a problem unsolvable sets the level of the ill-posed share; the drift is a change in that share under a fixed rule, and the more lenient open-weight judge finds a lower level but a similar drift ($+19.3$ against $+23.0$ points on the same problems).

Scope. Our loops use English grade-school math word problems and a policy-gradient trainer without a KL term; R-Zero's challenger writes competition-style problems, some partly in Chinese. The admission rules we ran as closed loops are R-Zero's agreement band, in our loop and in the official code, an abstention guard, a band recalibrated each round on the planted problems, and Absolute Zero's learnability rule computed on agreement; the other filters of Table 3 were applied offline to the band-trained loop, and in loops of their own they would select different data. We did not run other self-play methods such as SeRL (Fang et al., 2025) or self-questioning models (Chen et al., 2025). In domains where an executor or verifier certifies well-posedness, such as code, the band may behave differently (Pu et al., 2026). We trained models of up to 8B parameters for several rounds and ran one round at 14B; Phi-4-mini and the R-Zero replication have two seeds each, and Llama-3.2-3B, whose drift is small (2–6.5 points by round three), collapsed in three of five seeds. The R-Zero replication uses newer framework versions than the release pins, because the pinned versions do not run on our GPUs, and runs three of its five default iterations.

Attribution. The drift's own contribution to the solver's behaviour is estimated from one-round, same-size contrasts over five seeds, which do not resolve how much of it proposer-written rather than planted ill-posed problems contribute (§5); without planted premise-deleted problems the random subset also holds fewer intact planted problems than the solvable set (14–26 against 60–82; Appendix A).

The sharpening model and the probes. We fitted the class-conditional sharpening model after observing the fixed pool, refitted it after fixing a numerical bug, and part of the held-out pool had been scored when the fitting code was frozen; its held-out accuracy is therefore not a pre-registered prediction, unlike the class ordering in the two further models. The model approximates each problem's answer distribution from its agreement alone, so the values of its rates depend on how competing answers are modelled and only their ordering is robust; it never sharpens problems whose base mode is tied, and these problems carry its late-round plateau (Appendix D). The labels of readings, stated assumptions and gaps in §4.4 and §5 come from LLM labellers; two human annotators confirm the readings contrast and the rise in stated assumptions on a sample, and the labellers' label for agreeing answers proved lenient against exact matching (Appendix G). The annotators also found that the parser our probes first used read a number from verbal declines, about 30% of the round-5 solutions to unanswerable questions; we therefore re-ran the UMWP probes and the external sets with saved texts and report them under a stricter reading validated on their labels, which more than halves the rise in self-consistent answers (Appendix G), while the secondary premise-deleted probe keeps the first parser. Our behavioural rate measures commitment to one number, not stated certainty: after five rounds, 49–70% of the samples on unanswerable questions state an assumption or flag a gap, and 21–35% decline. Behaviour is measured with probes rather than in deployment, and the premise-deleted probe shares templates with the planted problems that enter training, which is why UMWP is our primary probe. The three round-5 seeds whose answers we compare share their initialisation and the probe's sampling seed, which may inflate their agreement (Appendix F).

Ethics Statement

This work studies a failure of label-free training that makes models answer questions they cannot answer. Answering a defective question with one

committed number can harm users wherever the defect goes unnoticed: a tutoring system that answers an ill-posed exercise tells a student that the exercise has a unique answer; a scientific or analytical tool that silently fills in a missing assumption returns a number whose basis the user never sees; and an assistant in a customer-facing setting may settle an incomplete or ambiguous request by guessing, without saying so. The risk is smaller when the answer states its assumption or declines, as half to two thirds of our trained solvers' samples do, and largest where only the final number is read. A test-time abstention option, which works in our experiments, may be insufficient on its own, because users may ignore the abstentions of a model that usually answers. We therefore recommend both measuring well-posedness independently of the model's own agreement during training and offering an explicit abstention option at test time. The released curricula can also be misused: their problems are written by models, and their labels come from language-model judges, with human labels for only 240 problems, so training or evaluating on them as if the labels were ground truth would propagate the judges' errors; they should not be used as ground truth for answerability without human review. On well-posed problems, the judges' false-positive rate is about 5%† on intact planted problems and higher in later rounds on the human-labelled problems (blind Opus: 2.9% at round one, 21–23.5% at rounds three and five; Figure 6), and the release marks the source of every label.

The training data consist of problems written by open models and of premise-deleted and intact GSM-Symbolic variants; ill-posedness labels come from language-model judges (Claude Opus, with Qwen2.5-32B-Instruct as a check) and, for 240 problems, from three external mathematicians (two PhD students and a postdoctoral researcher), who were paid USD 500 each for about five hours of work, consented to the use and anonymised release of their labels, and saw no personal data. The sample of Appendix G was labelled by two external annotators on the same terms of consent and data, for USD 1,000 each and 20.5 and 18.5 hours of work. The annotation was not reviewed by an ethics board. We use UMWP (CC BY-SA 4.0), SUM, MiP, GSM8K and MATH (MIT) and GSM-Symbolic, whose variants we release only as patches to respect its licence, and Qwen3, Qwen2.5, Phi-4-mini (MIT) and Llama 3.2 models under their licences.

Acknowledgments

We thank the three mathematicians who labelled the problems of Appendix B and the two annotators who labelled the sample of Appendix G. Claude (Anthropic), used through Claude Code, gave substantive assistance with the design of experiments and analyses, the code and its execution, the literature search, and the drafting and revision of the text; the author takes full responsibility for the content.

References

Abdelrahman Abouelenin, Atabak Ashfaq, Adam Atkinson, Hany Awadalla, Nguyen Bach, Jianmin Bao, Alon Benhaim, Martin Cai, Vishrav Chaudhary, Congcong Chen, et al. 2025. Phi-4-Mini technical report: Compact yet powerful multimodal language models via mixture-of-LoRAs. *Preprint*, arXiv:2503.01743.

Shivam Agarwal, Zimin Zhang, Lifan Yuan, Jiawei Han, and Hao Peng. 2025. The unreasonable effectiveness of entropy minimization in LLM reasoning. In *Advances in Neural Information Processing Systems*, volume 38, pages 107150–107180. Curran Associates, Inc.

Luke Bailey, Kaiyue Wen, Kefan Dong, Tatsunori Hashimoto, and Tengyu Ma. 2026. Scaling self-play with self-guidance. In *Third Conference on Language Modeling*.

Nuel D. Belnap, Jr. and Thomas B. Steel, Jr. 1976. *The Logic of Questions and Answers*. Yale University Press, New Haven, CT.

Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. 2009. Curriculum learning. In *Proceedings of the 26th Annual International Conference on Machine Learning*, pages 41–48. ACM.

Yuri Burda, Harri Edwards, Deepak Pathak, Amos Storkey, Trevor Darrell, and Alexei A. Efros. 2019. Large-scale study of curiosity-driven learning. In *The Seventh International Conference on Learning Representations*.

Xujun Che, Yuchen Yuan, Weida Zhao, and Chenyang Yu. 2026. Abstention as an action can kill both the reward gradient and the KL anchor: Collapse law and repair for error-penalized reinforcement learning. *Preprint*, arXiv:2608.00301.

Lili Chen, Mihir Prabhudesai, Katerina Fragkiadaki, Hao Liu, and Deepak Pathak. 2025. Self-questioning language models. *Preprint*, arXiv:2508.03682.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *Preprint*, arXiv:2110.14168.

Jeremy Cole, Michael Zhang, Daniel Gillick, Julian Eisenschlos, Bhuwan Dhingra, and Jacob Eisenstein. 2023. Selectively answering ambiguous questions. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 530–543, Singapore. Association for Computational Linguistics.

Aditya Cowsik, Kfir Dolev, Michael Y. Li, G. Bruno De Luca, Nourya Cohen, Noah D. Goodman, and Yoav Levine. 2026. Self-play pretraining with zero data. *Preprint*, arXiv:2609.30063.

Michael Dennis, Natasha Jaques, Eugene Vinitsky, Alexandre Bayen, Stuart Russell, Andrew Critch, and Sergey Levine. 2020. Emergent complexity and zero-shot transfer via unsupervised environment design. In *Advances in Neural Information Processing Systems*, volume 33, pages 13049–13061. Curran Associates, Inc.

Jacob Dineen, Aswin RRV, Zhikun Xu, and Ben Zhou. 2026. Vocabulary dropout for curriculum diversity in LLM co-evolution. In *Third Conference on Language Modeling*.

Chenrui Fan, Ming Li, Lichao Sun, and Tianyi Zhou. 2025. Missing premise exacerbates overthinking: Are reasoning models losing critical thinking skill? In *Second Conference on Language Modeling*.

Wenkai Fang, Shunyu Liu, Yang Zhou, Kongcheng Zhang, Tongya Zheng, Kaixuan Chen, Mingli Song, and Dacheng Tao. 2025. SeRL: Self-play reinforcement learning for large language models with limited data. In *Advances in Neural Information Processing Systems*, volume 38, pages 103706–103738. Curran Associates, Inc.

Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. 2024. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. 2024. The Llama 3 herd of models. *Preprint*, arXiv:2407.21783.

Renjie Gu, Jiaxu Li, Yihao Wang, Yun Yue, Hansong Xiao, Yefei Chen, Yuan Wang, Chunxiao Guo, Pei Wei, Jinjie Gu, and Yixin Cao. 2026. Bridging the detection-to-abstention gap in reasoning models under insufficient information. *Preprint*, arXiv:2605.28070.

Bingxiang He, Yuxin Zuo, Zeyuan Liu, Shangziqi Zhao, Zixuan Fu, Junlin Yang, Cheng Qian, Kaiyan Zhang, Yuchen Fan, Ganqu Cui, Xiusi Chen, Youbang Sun, Xingtai Lv, Xuekai Zhu, Li Sheng, Ran Li, Huan-ang Gao, Yuchen Zhang, Lifan Yuan, and 3 others. 2026. How far can unsupervised RLVR scale LLM training? In *The Fourteenth International Conference on Learning Representations*.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. 2021. Measuring mathematical problem solving with the MATH dataset. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, volume 1.

Bairu Hou, Yujian Liu, Kaizhi Qian, Jacob Andreas, Shiyu Chang, and Yang Zhang. 2024. Decomposing uncertainty for large language models through input clarification ensembling. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 19023–19042. PMLR.

Neil Houlsby, Ferenc Huszár, Zoubin Ghahramani, and Máté Lengyel. 2011. Bayesian active learning for classification and preference learning. *Preprint*, arXiv:1112.5745.

Shengding Hu, Yifan Luo, Huadong Wang, Xingyi Cheng, Zhiyuan Liu, and Maosong Sun. 2023. Won't get fooled again: Answering questions with false premises. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5626–5643, Toronto, Canada. Association for Computational Linguistics.

Audrey Huang, Adam Block, Dylan J. Foster, Dhruv Rohatgi, Cyril Zhang, Max Simchowitz, Jordan T. Ash, and Akshay Krishnamurthy. 2025. Self-improvement in language models: The sharpening mechanism. In *The Thirteenth International Conference on Learning Representations*.

Chengsong Huang, Haolin Liu, Tong Zheng, Runpeng Dai, Langlin Huang, Jinyuan Li, Zongxia Li, Zhepei Wei, Yu Meng, and Jiaxin Huang. 2026a. G-Zero: Self-play for open-ended generation from zero data. In *Advances in Neural Information Processing Systems*. To appear; arXiv:2605.09959.

Chengsong Huang, Wenhao Yu, Xiaoyang Wang, Hongming Zhang, Zongxia Li, Ruosen Li, Jiaxin Huang, Haitao Mi, and Dong Yu. 2026b. R-Zero: Self-evolving reasoning LLM from zero data. In *The Fourteenth International Conference on Learning Representations*.

Jiaxin Huang, Shixiang Gu, Le Hou, Yuexin Wu, Xuezhi Wang, Hongkun Yu, and Jiawei Han. 2023. Large language models can self-improve. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 1051–1068, Singapore. Association for Computational Linguistics.

Eyke Hüllermeier and Willem Waegeman. 2021. Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods. *Machine Learning*, 110(3):457–506.

Minqi Jiang, Edward Grefenstette, and Tim Rocktäschel. 2021. Prioritized level replay. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 4940–4950. PMLR.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislav Fort, and 17 others. 2022. Language models (mostly) know what they know. *Preprint*, arXiv:2207.05221.

Amita Kamath, Robin Jia, and Percy Liang. 2020. Selective question answering under domain shift. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 5684–5696, Online. Association for Computational Linguistics.

Siddharth Karamcheti, Ranjay Krishna, Li Fei-Fei, and Christopher Manning. 2021. Mind your outliers! Investigating the negative impact of outliers on active learning for visual question answering. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7265–7281, Online. Association for Computational Linguistics.

Lauri Karttunen. 1974. Presupposition and linguistic context. *Theoretical Linguistics*, 1(1–3):181–194.

Lauri Karttunen. 1977. Syntax and semantics of questions. *Linguistics and Philosophy*, 1(1):3–44.

Alex Kendall and Yarin Gal. 2017. What uncertainties do we need in Bayesian deep learning for computer vision? In *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc.

Najoung Kim, Phu Mon Htut, Samuel R. Bowman, and Jackson Petty. 2023. $(QA)^2$: Question answering with questionable assumptions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8466–8487, Toronto, Canada. Association for Computational Linguistics.

Najoung Kim, Ellie Pavlick, Burcu Karagol Ayan, and Deepak Ramachandran. 2021. Which linguist invented the lightbulb? presupposition verification for question-answering. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 3932–3945, Online. Association for Computational Linguistics.

Polina Kirichenko, Mark Ibrahim, Kamalika Chaudhuri, and Samuel J. Bell. 2025. AbstentionBench: Reasoning LLMs fail on unanswerable questions. In *Advances in Neural Information Processing Systems*, volume 38. Curran Associates, Inc. Datasets and Benchmarks Track.

Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2023. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *The Eleventh International Conference on Learning Representations*.

Wai-Chung Kwan, Joshua Ong Jun Leang, Pavlos Vougiouklis, Jeff Z. Pan, Marco Valentino, and Pasquale Minervini. 2025. OpenSIR: Open-ended self-improving reasoner. *Preprint*, arXiv:2511.00602.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. 2023. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP '23)*, pages 611–626. ACM.

David Lewis. 1979. Scorekeeping in a language game. *Journal of Philosophical Logic*, 8(1):339–359.

David K. Lewis. 1969. *Convention: A Philosophical Study*. Harvard University Press, Cambridge, MA.

Belinda Z. Li, Been Kim, and Zi Wang. 2025. QuestBench: Can LLMs ask the right question to acquire information in reasoning tasks? In *Advances in Neural Information Processing Systems*, volume 38. Curran Associates, Inc. Datasets and Benchmarks Track.

Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2024. Let's verify step by step. In *The Twelfth International Conference on Learning Representations*.

Alisa Liu, Zhaofeng Wu, Julian Michael, Alane Suhr, Peter West, Alexander Koller, Swabha Swayamdipta, Noah Smith, and Yejin Choi. 2023. We're afraid language models aren't modeling ambiguity. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 790–807, Singapore. Association for Computational Linguistics.

Sewon Min, Julian Michael, Hannaneh Hajishirzi, and Luke Zettlemoyer. 2020. AmbigQA: Answering ambiguous open-domain questions. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 5783–5797, Online. Association for Computational Linguistics.

Sören Mindermann, Jan M. Brauner, Muhammed T. Razzak, Mrinank Sharma, Andreas Kirsch, Winnie Xu, Benedikt Höltgen, Aidan N. Gomez, Adrien Morisot, Sebastian Farquhar, and Yarin Gal. 2022. Prioritized training on points that are learnable, worth learning, and not yet learnt. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 15630–15649. PMLR.

Iman Mirzadeh, Keivan Alizadeh-Vahid, Hooman Shahrokhi, Oncel Tuzel, Samy Bengio, and Mehrdad Farajtabar. 2025. GSM-Symbolic: Understanding the limitations of mathematical reasoning in large language models. In *The Thirteenth International Conference on Learning Representations*.

Mihir Prabhudesai, Lili Chen, Alex Ippoliti, Katerina Fragkiadaki, Hao Liu, and Deepak Pathak. 2025. Maximizing confidence alone improves reasoning. *Preprint*, arXiv:2505.22660.

Archiki Prasad, Weizhe Yuan, Richard Yuanzhe Pang, Jing Xu, Maryam Fazel-Zarandi, Mohit Bansal, Sainbayar Sukhbaatar, Jason E. Weston, and Jane Yu. 2025. Self-consistency preference optimization. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 49737–49751. PMLR.

Sophia Xiao Pu, Zhaotian Weng, Chengzhi Liu, Jayanth Srinivasa, Gaowen Liu, William Yang Wang, and Xin Eric Wang. 2026. Survive or collapse: The asymmetric roles of data gating and reward grounding in self-play RL. In *Advances in Neural Information Processing Systems*. To appear; arXiv:2605.22217.

Qwen Team. 2024. Qwen2.5 technical report. *Preprint*, arXiv:2412.15115.

Pranav Rajpurkar, Robin Jia, and Percy Liang. 2018. Know what you don't know: Unanswerable questions for SQuAD. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 784–789, Melbourne, Australia. Association for Computational Linguistics.

Willem Röpke, Samuel Coward, Andrei Lupu, Thomas Foster, Tim Rocktäschel, and Jakob Foerster. 2026. DéjàQ: Open-ended evolution of diverse, learnable and verifiable problems. *Preprint*, arXiv:2601.01931.

Thomas C. Schelling. 1960. *The Strategy of Conflict*. Harvard University Press, Cambridge, MA.

Sheikh Shafayat, Fahim Tajwar, Ruslan Salakhutdinov, Jeff Schneider, and Andrea Zanette. 2025. Can large reasoning models self-train? *Preprint*, arXiv:2505.21444.

Rulin Shao, Shuyue Stella Li, Rui Xin, Scott Geng, Yiping Wang, Sewoong Oh, Simon Shaolei Du, Nathan Lambert, Sewon Min, Ranjay Krishna, Yulia Tsvetkov, Hannaneh Hajishirzi, Pang Wei Koh, and Luke Zettlemoyer. 2026. Spurious rewards: Rethinking training signals in RLVR. In *Proceedings of the 43rd International Conference on Machine Learning*.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. 2024. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *Preprint*, arXiv:2402.03300.

Linxin Song, Taiwei Shi, and Jieyu Zhao. 2025. The hallucination tax of reinforcement finetuning. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 2105–2120, Suzhou, China. Association for Computational Linguistics.

YuHong Sun, Zhangyue Yin, Qipeng Guo, Jiawen Wu, Xipeng Qiu, and Hui Zhao. 2024. Benchmarking hallucination in large language models based on unanswerable math word problem. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 2178–2188, Torino, Italia. ELRA and ICCL.

Shobhita Sundaram, John Quan, Ariel Kwiatkowski, Kartik Ahuja, Yann Ollivier, and Julia Kempe. 2026. Teaching models to teach themselves: Reasoning at the edge of learnability. In *Proceedings of the 43rd International Conference on Machine Learning*.

Swabha Swayamdipta, Roy Schwartz, Nicholas Lourie, Yizhong Wang, Hannaneh Hajishirzi, Noah A. Smith, and Yejin Choi. 2020. Dataset cartography: Mapping and diagnosing datasets with training dynamics. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9275–9293, Online. Association for Computational Linguistics.

Yongqi Tong, Zhenyu Zhang, Zimi Liu, Kewei Fu, Mingli Song, Haofei Zhang, Junshao Zhang, Hong Zhu, Jiang-Ming Yang, Xin Zhang, and Jianshe Li. 2026. Ask, condition or abstain: Reinforcement learning for missing-premise reasoning. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*. To appear; arXiv:2608.16554.

Emiel van Miltenburg, Chris van der Lee, and Emiel Krahmer. 2021. Preregistering NLP research. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 613–623, Online. Association for Computational Linguistics.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations*.

Bingbing Wen, Jihan Yao, Shangbin Feng, Chenjun Xu, Yulia Tsvetkov, Bill Howe, and Lucy Lu Wang. 2025. Know your limits: A survey of abstention in large language models. *Transactions of the Association for Computational Linguistics*, 13:529–556.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. 2025. Qwen3 technical report. *Preprint*, arXiv:2505.09388.

Ziyi Yang, Weizhou Shen, Chenliang Li, Ruijun Chen, Fanqi Wan, Ming Yan, Xiaojun Quan, and Fei Huang. 2026. SPELL: Self-play reinforcement learning for evolving long-context language models. In *The Fourteenth International Conference on Learning Representations*.

Zhangyue Yin, Qiushi Sun, Qipeng Guo, Jiawen Wu, Xipeng Qiu, and Xuanjing Huang. 2023. Do large language models know what they don't know? In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 8653–8665, Toronto, Canada. Association for Computational Linguistics.

Jing Yu, Shengchao Chen, and Yiyun Tan. 2026. The chase is the curriculum, the capture anchors the credit: Pursuit-evasion self-play for zero-data LLM reasoning. *Preprint*, arXiv:2608.21871.

Qiying Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian Fan, Gaohong Liu, Juncai Liu, Lingjun Liu, Xin Liu, Haibin Lin, Zhiqi Lin, Bole Ma, Guangming Sheng, Yuxuan Tong, Chi Zhang, Mofan Zhang, and 17 others. 2025a. DAPO: An open-source LLM reinforcement learning system at scale. In *Advances in Neural Information Processing Systems*, volume 38, pages 113222–113244. Curran Associates, Inc.

Wenhao Yu, Zhenwen Liang, Chengsong Huang, Kishan Panaganti, Tianqing Fang, Haitao Mi, and Dong Yu. 2025b. Guided self-evolving LLMs with minimal human supervision. *Preprint*, arXiv:2512.02472.

Xinyan Yu, Sewon Min, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. CREPE: Open-domain question answering with false presuppositions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10457–10480, Toronto, Canada. Association for Computational Linguistics.

Hanning Zhang, Shizhe Diao, Yong Lin, Yi Fung, Qing Lian, Xingyao Wang, Yangyi Chen, Heng Ji, and Tong Zhang. 2024. R-Tuning: Instructing large language models to say 'I don't know'. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 7113–7139, Mexico City, Mexico. Association for Computational Linguistics.

Qingyang Zhang, Haitao Wu, Changqing Zhang, Peilin Zhao, and Yatao Bian. 2025. Right question is already half the answer: Fully unsupervised LLM reasoning incentivization. In *Advances in Neural Information Processing Systems*, volume 38, pages 67345–67372. Curran Associates, Inc.

Andrew Zhao, Yiran Wu, Tong Wu, Quentin Xu, Yang Yue, Matthieu Lin, Shenzhi Wang, Qingyun Wu, Zilong Zheng, and Gao Huang. 2025. Absolute Zero: Reinforced self-play reasoning with zero data. In *Advances in Neural Information Processing Systems*, volume 38, pages 105816–105879. Curran Associates, Inc.

Xuandong Zhao, Zhewei Kang, Aosong Feng, Sergey Levine, and Dawn Song. 2026. Learning to reason without external rewards. In *The Fourteenth International Conference on Learning Representations*.

Chenyu Zhou. 2026. More convincing, not more correct: Self-play reward hacking of reference-free LLM judges. *Preprint*, arXiv:2607.05904.

Yuxin Zuo, Kaiyan Zhang, Li Sheng, Shang Qu, Ganqu Cui, Xuekai Zhu, Haozhan Li, Yuchen Zhang, Xinwei Long, Ermo Hua, Biqing Qi, Youbang Sun, Zhiyuan Ma, Lifan Yuan, Ning Ding, and Bowen Zhou. 2025. TTRL: Test-time reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 38, pages 131459–131483. Curran Associates, Inc.

A Setup Details

Answer parsing in the loop (filter reading). Generation by a base model stops at a new "Question:" or "Problem:" line. We take the last boxed expression; otherwise an explicit "Answer: unknown" counts as abstention; otherwise the text after "####"; otherwise the last number. Numbers are normalised before comparison. Behavioural numbers use the strict reading of §2 instead (Appendix G).

Planted problems. Admitted premise-deleted problems make up 11.5–22.8% of the training sets of the full curricula (Qwen3-4B, Qwen3-8B and Phi-4-mini; 45 runs) and admitted intact ones 6.3–17.7%; from round two on, every run admits more deleted than intact ones. None of the 23,241 judge labels concerns a planted problem.

Training-set size and passes. Every round trains for 60 steps of 32 problems, drawn by cycling through a reshuffled training set, so a set of n problems is passed over $1{,}920/n$ times, with eight rollouts per pass. The admitted set shrinks as the curriculum drifts, and the number of passes grows (Figure 11 in Appendix F).

Controls and variants. The *solvable-only* curriculum trains on the admitted proposals that the judges, labelling every admitted proposal, call solvable, plus the admitted intact planted problems, and no deleted ones. The *size-matched random loop* admits, in every round, a random subset of the band of the size the abstention guard admits (62–69% of the full curriculum at round three). The *same-size contrast* trains one round from the round-2 solver either on the solvable proposals of the round-3 admitted set plus its intact planted problems, or on a random subset of that set of the same size (148–177 problems, of which 20–39 are premise-deleted). The *same-size contrast without planted problems*, registered before any of its new runs existed, replaces in place each premise-deleted problem of

Proposals	1,200 requests per round (40 topic prompts $\times$ 30), T=1.0, top-p 0.95, $\leq$300 tokens; kept if $\geq$40 characters, with a digit and a question mark, and not a duplicate
Planted problems	200 premise-deleted and 200 intact GSM-Symbolic variants per round, disjoint from the evaluation probe
Admission	G=8 samples, T=1.0, $\leq$1,024 tokens; p = share of the most frequent parsed answer (unparsed samples count in the denominator); admit if $0.25 \leq p \leq 0.75$
Training	60 steps per round, 32 problems $\times$ 8 rollouts; reward 1 for matching the group majority; advantage = reward $-$ group mean, not divided by the group's standard deviation; one on-policy update per batch, so no ratio clipping is needed; no KL term; learning rate 10^{-6} (constant); gradient clip 1.0; full-parameter, bf16 weights with fp32 master
Optimiser	8-bit AdamW on GPU (Qwen3-4B, Phi-4-mini, Llama-3.2-3B); fp32 AdamW on CPU (Qwen3-8B, Qwen3-14B)
Labels	200 admitted proposals per round, 8 judges $\times$ 25 (10 $\times$ 20 for the first four round-1 samples)
UMWP probe	600 unanswerable + 600 answerable originals, 8 samples, T=1.0
Accuracy	greedy; MATH-500, 1,000 further MATH test problems, 500 GSM8K test problems
Compute	8 NVIDIA RTX PRO 5000 (72 GB) per host, 3 hosts; about 1 h (4B) to 3 h (14B) of training per round; about 1,400 GPU-hours in total (1,300–1,600), 435 of them for the R-Zero replication
Software	vLLM 0.11.0 (Kwon et al., 2023) for sampling, PyTorch 2.8.0, transformers 4.57.1, bitsandbytes 0.48.1 8-bit AdamW, our own policy-gradient trainer; no hyper-parameter search

Table 4: Loop settings. Base models see the problem followed by "Let's think step by step and output the final answer within \boxed{}"; instruction-tuned models see the same text in their chat template. The supplementary material lists the 40 topic prompts and the proposer's instruction template.

that random subset (20–39 per seed) with a random admitted proposal not already in it, leaving the size, the order of the other problems and every setting unchanged; its runs were trained three days after the others, and GPU sampling is not bitwise reproducible. Of its problems, 103–117 are ill-posed proposals (79–95 in the random subset) and 14–26 intact planted problems, against none and 60–82 in the solvable set. Putting the planted premise-deleted problems back in place of these proposals changes the rate by +0.3 points on average (95% CI $[-3.0, 3.6]$; higher in three of five seeds; answers read as in §2); this and the difference without them add up to the same-size contrast by construction. With the abstention option, one round on the solvable problems, on the random subset and on the subset without planted problems abstains on 70, 71 and 68% of the unanswerable UMWP questions and on 2, 2 and 2% of the answerable ones (seed means, five seeds). The Absolute-Zero-style band admits $0 < p < 1$. The co-evolving proposer is fine-tuned after each round for two epochs on the proposals the band admitted (learning rate 10^{-5}).

B Judges and Their Validation

Judges. Each judge is a Claude Opus agent (Opus 5 series, September 2026; the blind re-judging and the R-Zero labels used `claude-opus-5-5`) with a fresh context and shell access, which fetches its batch of problems with one command and may check computations locally. It receives the following definitions (verbatim): "These were written by a [model] language model and ADMITTED by a self-play filter. For EACH question decide: SOLVABLE – the text determines a unique numeric answer (state it briefly in the note); UNSOLVABLE – a needed quantity/relation is missing, or the text is self-contradictory, so no unique answer exists; AMBIGUOUS – two reasonable readings give different answers. Be strict: a default assumption you would have to invent makes it UNSOLVABLE. Confidence 0-1, note $\leq$ 20 words." The framing sentence is the same in every round.

Validation. Figure 5 collects the checks. On UMWP, the 200 unanswerable questions were stratified by source and category and the 200 answerable ones by source. On 200 planted problems of known status, the judge labels 92 of 100 deletions and 12 of 100 intact problems ill-posed; the notes of all eight deletions judged solvable give an

answer that differs from the original's, i.e. these deletions left the problem solvable. An LLM reader compared the 12 intact problems with their reference answers and with the judge's labels for other instances of the same template[†]: 7 are genuine defects of the instance (one needs 112% of a working day, two assume four weeks in April or July, and four are ambiguous, three of them from one template whose premise sentence, placed after the question, reads as a final total), and 5 are judge false positives, four of which contradict the judge's solvable label for another instance of the same template. The false-positive rate on well-posed intact problems is thus about 5% (5 of 93). Re-judging two samples with new judges and reversed order reproduces the ill-posed share within 2.5 points; disagreements concentrate on the boundary between unsolvable and ambiguous, which is why we report their sum.

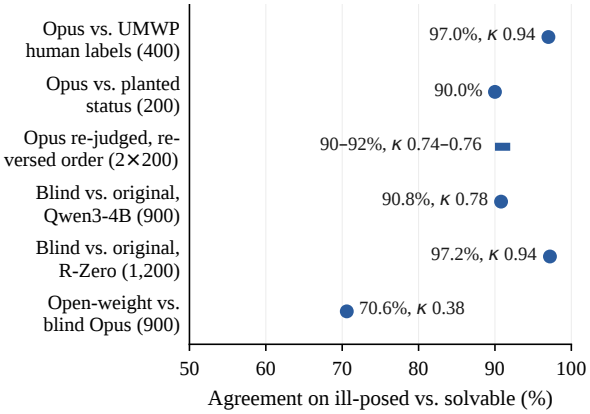


Figure 5: Agreement on ill-posed vs. solvable across judges and against human labels, with Cohen's κ (number of problems in parentheses).

Human annotation. We sampled 240 of the 900 problems re-judged blind (Qwen3-4B, seeds 0–2, rounds 1, 3 and 5; 80 per round, 26–27 per seed and round), shuffled them under neutral identifiers and gave them to two external annotators, PhD students in mathematics (fourth and third year), with the instruction: "Label each question with exactly one of: S (SOLVABLE): the text determines a unique answer without inventing any assumption. U (UNSOLVABLE): a needed quantity or relation is missing, or the text is self-contradictory, so no unique answer exists. A (AMBIGUOUS): two reasonable readings give different answers. Be strict: if you would have to invent a default assumption to get a number, label U." They worked independently, without LLM assistance and without having read this paper or seen model outputs; a postdoctoral mathematician adjudicated the 17 disagreements, seeing only the two labels, not the round or any judge label. Each was paid USD 500 for about five hours. The analysis plan and script were registered after the annotation but before the labels were joined with the round and judge information (Appendix C, deviation 14). Figure 6 shows the results; on the same problems, blind Opus agrees with the annotators at κ=0.89 and measures a somewhat larger drift on the same problems (+26.2; paired difference +6.2, CI $[0.0, 12.5]$). The confidence intervals come from an item bootstrap within the three runs. The estimated rise is positive in every seed (+14.8, +22.2 and +23.1 points; with 26–27 problems per cell, each per-seed interval reaches 0) and under each annotator's labels alone (+18.8 and +17.5, both intervals above 0); a logistic trend over rounds gives an odds ratio of 1.28 per round (95% CI $[1.07, 1.52]$). Corrected for the judge's per-round error rates estimated from these 240 problems, the blind Opus drift on all 900 problems is +17.2 (Rogan–Gladen, CI $[2.8, 28.1]$) or +18.4 points (two-phase estimator, CI $[9.1, 27.3]$)[†], against +24.7 uncorrected.

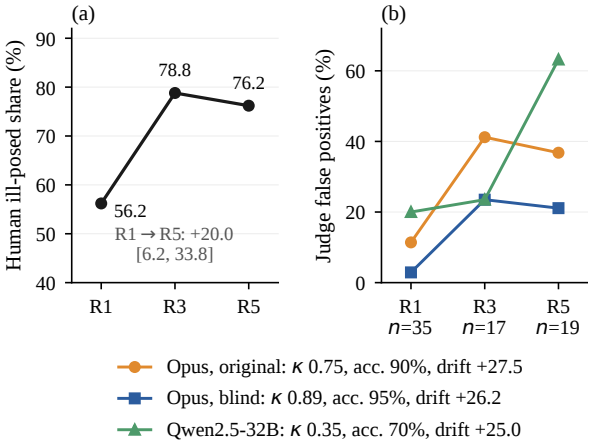


Figure 6: Human annotation of 240 Qwen3-4B problems. (a) Ill-posed share under the final human labels, with the change from round one to five and its 95% CI. (b) False-positive rate of each judge: the share of human-solvable problems (n per round) that it calls ill-posed; legend: binary κ and accuracy against the final human label, and the drift the judge measures on the same 240 problems. The annotators agree at κ=0.87 (three classes, CI $[0.81, 0.92]$) and 0.91 (ill-posed vs. solvable, $[0.85, 0.96]$).

Blind re-judging. We re-judged blind the labelled samples of Qwen3-4B (seeds 0–2, rounds 1, 3 and 5) and of R-Zero. We shuffled the problems

under neutral keys into one file whose name carried no information about their origin; each judge fetched only its own lines, and we checked afterwards which files each judge had read. In the R-Zero re-judging (49 judge batches), one batch of 25 was discarded and re-judged, as the pre-registered protocol required, because its judge had read the output of its own background computation, which the check did not permit.

C Pre-Registration and Outcomes

Procedure. We pre-registered eleven sets of predictions. For each, we wrote the predictions, thresholds, analysis scripts and the conditions for stopping a run before generating the data, and appended their SHA-256 hashes, with timestamps, to an append-only log that we maintain ourselves; from the third set on, most analysis scripts were reviewed by LLM agents before their first run. Figure 7 summarises the outcomes, Table 5 maps each claim of the abstract to its evidence and registration status, and the supplementary material lists all 136 predictions (S1–S136) with their thresholds and outcomes. Its software and data archives also contain the registration log with the registered documents and scripts of every study reported here and a script that recomputes their hashes, all prompts (including the corrected abstention prompt), the item identifiers, seeds and analysis scripts, label tables that give the source of every label, and our patch to the R-Zero code.

Predictions that did not hold. *Remedies:* the abstention guard was predicted to keep the admitted set at most 35% ill-posed with a rise under 10 points, and the loop's unsolvable share under 30% (neither held; §6); a rule admitting unanimous problems was predicted to drift by less than 10 points on the held-out pool (it drifted by 11.9 points); a stronger proposer and harder refills (solved problems rewritten into harder ones) were predicted to escape the drift (they did not); a co-evolving proposer was predicted to write more ill-posed problems and to cost accuracy (it wrote fewer and cost none); a band recalibrated each round on the planted problems was predicted to stop the drift (it delayed it); rewarding unknown was predicted to lower self-consistent answering and raise abstention in every seed and to stay stable through round five (each held in some seeds only), and a triple weight to avoid refusal collapse (it collapsed); offering unknown without reward was predicted to raise

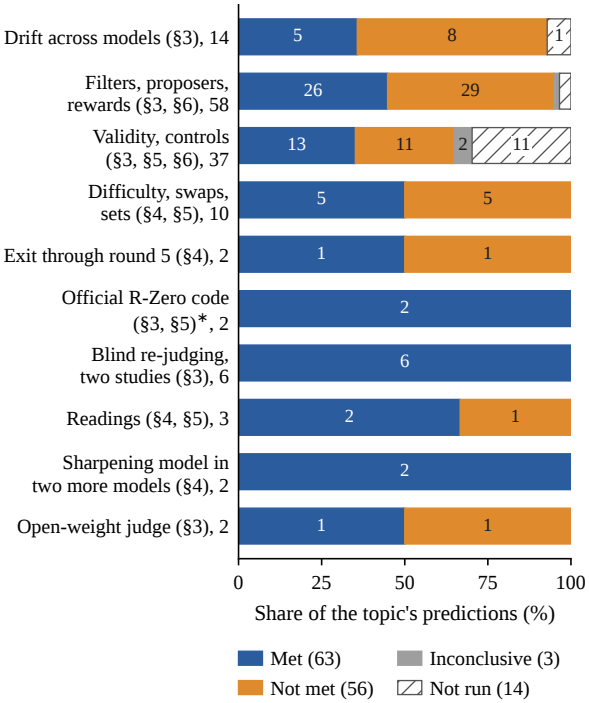


Figure 7: Outcomes of the pre-registered predictions by topic (number of predictions after each topic; counts in the bars). Met / not met: the outcome meets or misses the pre-registered criterion. The two unlabelled segments of filters, proposers and rewards are 1 inconclusive and 2 not run. *Both R-Zero predictions are met by the pre-registered analysis script, which we ran once all computation had ended, before the times fixed for it; the runs at those times give byte-identical output (deviation 9).

Claim of the abstract	Evidence	Registration and outcome
The ill-posed share of admitted problems rises by 11.5–27 points within two rounds	§3, Table 1	Unsolvable share: S3 (Qwen3-4B/8B) and S26 (Phi-4-mini) met; S38 (Qwen3-14B, $+6.0$ against $+10$) not met; Llama-3.2-3B drift (S2, S4) not met. The ill-posed share is reported alongside
…also under blind re-judging, an open-weight judge and expert human annotation	§3, Table 1, Figure 6	S124–S129 met; S135 met; S136 ($\kappa \geq 0.40$) not met; human annotation: rises (registered after annotation, before unblinding)
The official R-Zero code drifts, mostly through its challenger	§3, App. E	S122 met; challenger and selection shares from a pre-registered analysis with a pre-stated interpretation (raw share rises; selection positive; the band's own part not established)
A fixed pool isolates the filter	§4.1, Figure 3	Fixed pool†; held-out pool S29, S31 met, S30 not met
Solvable problems sharpen out; problems with several readings stay	§4.2–4.4	Sharpening fit and held-out error†; class ordering S133, S134 met; readings S130, S132 met, S131 not met; human check of the readings labels: criterion met (App. G); retention against hard answerable questions S111 not met; exits†
Self-consistent answers rise while accuracy plateaus	§5, Figure 2b	Registered, first parser on the original runs: S66 met (28.7% to 58.7–75.8%), equal accuracy across curricula S9, S41 met, a predicted accuracy cost (S5) not met. Strict reading on the new runs (exploratory†): 28% to 41–48%, S66's comparison holds in every seed (App. G)
Most of the rise is sharpening; what the ill-posed problems add is not established	§5, Table 2	Solvable-only curriculum lower: S65 met; same-size UMWP contrast†; its premise-deleted version S114 not met per seed (pooled test met), S115 (Phi-4-mini) not met; planted-free contrast, registered separately: no difference detected (all registered outcomes: first parser). Strict reading†: no contribution detected at matched size
No label-free admission rule we tested stops the drift	§6, Table 3	Held-out rules S29 met, S30 not met; abstention guard S15, S16 not met; further fixed-pool rules†
Offering abstention at test time works	§5–6, Figure 12	S119 met; S76 not met in the opposite direction; loss of separability (S79) not met; three predictions on the corrected probe met; held-out calibration†
136 predictions, every outcome reported	App. C	63 met, 56 not met, 3 inconclusive, 14 not run

Table 5: Claim-status map. S-numbers refer to the list of predictions in the supplementary material; †: not pre-registered.

abstention (it lowered it below the base model's rate). *Controls:* the guard was predicted to beat a size-matched random loop in at least two of three seeds, and the size of the admitted set was predicted not to matter (neither held); the one-round same-size contrast (§5) was predicted to lower self-consistent answering on premise-deleted problems by 4 points in most seeds (the pooled effect is significant, but the per-seed margin is met in two of five seeds) and in Phi-4-mini on UMWP (met in one of two seeds). *Mechanism:* unanswerable problems were predicted to be retained more than hard answerable ones (§4.3), also on premise-deleted problems (five of ten lineages); differential exit was predicted to keep widening through round five (one of three seeds); explicit flagging of missing information was predicted to fall (it rose; §5); the agreement signal was predicted to lose its ability to separate answerable from unanswerable questions (it did not; §6). *Scope:* the drift predictions for Llama-3.2-3B did not hold; the cost predicted by round three included an accuracy drop, which did not occur; Qwen3-8B was predicted to match Qwen3-4B within 0.10 on two statistics; the unsolvable-only share of the 14B round rose by 6 points against a predicted 10; the guard's answerable accuracy on UMWP fell by more than 2 points in one seed; the open-weight judge's agreement with Opus was κ=0.38 against a predicted 0.40; the margin on the external unanswerable sets was met on one of three sets.

Further pre-registered analyses. Six analyses were registered with a pre-stated interpretation for each possible outcome instead of pass/fail predictions, four before any of their data existed, the same-size contrast without planted problems before any of its new runs existed and the human annotation after labelling but before unblinding (deviation 14); they are not counted among the 136: the human annotation (the human ill-posed share rises from round one to five; Appendix B), the blind judging of R-Zero's raw challenger output (the raw share rises, and downstream selection adds to it; §3), R-Zero's benchmark accuracy (a gain in one of two seeds in R-Zero's format; Appendix E), a verifier on top of the band (the level falls, the drift persists, and the change in odds is consistent with a fixed miss rate; §6) the same-size contrast without planted problems (not resolved at five seeds; §5) and the human check of the mechanism labels (both criteria met; Appendix G).

Not run. A fourth model family (OLMo-2), a 14B proposer, the first seed of the 8B guard lineage and its probe evaluations, three predictions on MiP, a follow-up to a reward variant that we stopped, as pre-registered, when its completed seed abstained on 67% of intact problems, a loop variant that was to run only if that follow-up succeeded, and a reward variant whose preliminary study did not meet its criterion.

Deviations. (1) The solvable-only predictions were written after two of four seeds had been seen, as the pre-registration states; only the other two seeds test them. (2) The UMWP replication of that contrast was pre-registered after the premise-deleted result was known, the exit prediction through round five after its round-3 terms were known, and one continued-loop comparison after the frontier values it uses had been computed. (3) Three outcomes, and one clause of two further predictions, were computed by arithmetic from hashed outputs rather than by a hashed script. (4) The open-weight judge's output file was hashed together with, rather than before, its analysis script. (5) One validity analysis used Monte Carlo subsampling (2,000 draws per item) instead of the planned exact computation. (6) One control curriculum could not be matched exactly in composition (6–25 unsolvable problems short). (7) One Llama seed was added by an entry in the log rather than in the pre-registration text, and in one study three evaluations, and the third-seed evaluations of three further predictions, finished after the planned cut-off; without them no outcome changes. (8) We refitted the sharpening model after finding a numerical bug, when the original held-out error (3.0 points) was already known. (9) The R-Zero analyses were run once all computation had ended, before the times fixed in their pre-registration; the runs at those times (72 and 84 hours after registration) give byte-identical output (Figure 7). (10) Qwen3-8B and 14B used fp32 AdamW on the CPU rather than 8-bit AdamW. (11) The abstention-prompt probes for UMWP and the three external sets first contained a formatting error in the instruction; the `unknown` option itself was intact. We re-ran the 30 checkpoints whose abstention numbers we report with the corrected instruction, under a separate pre-registration of three further predictions (not counted among the 136), and report abstention numbers only from this run. Abstention rates changed by -3 to $+9$ points (most for the base

model) and self-consistent-answer rates with the option by -7 to $+2$. The three new predictions hold, and the earlier predictions that use this probe reach the same outcomes on the corrected data, except one that needs Phi-4-mini, which we did not re-run. (12) In the blind judging of R-Zero's raw challenger output, we re-judged every batch whose judge had read files that could reveal an item's origin. From the second pass on, a judge's computations in its own scratch folder, and from the third pass on, the output of its own background computations, were no longer counted; the first pass keeps the original check. (13) In the benchmark comparison, the base model's MATH-500 accuracy under our prompt was 2.4 points below an earlier evaluation of the same model, beyond the pre-stated tolerance of 2 points. (14) The analysis plan and script for the human annotation were written and registered after the annotation was complete but before the labels were joined with the round, seed and judge information; the annotators were blind to all three. (15) The human check of the mechanism labels (Appendix G) was first registered with a packet of 40 units and 120 solutions; the label files received for it could not be established as independent annotations and were set aside without analysis, and the check was re-registered, before any of its labels existed, on 184 units and 600 solutions that the first packet did not contain. (16) The probes behind the pre-registered behavioural predictions were scored with a parser that read numbers from verbal declines (Appendix G). Their outcomes, in Table 5 and above, are those of the original runs under that parser. The UMWP and external-set numbers of the main text come from a re-run with saved texts under the strict reading of §2, planned and logged after the human check exposed the error and before any new output was read; they are exploratory corrections, not confirmatory tests. Under the strict reading, the comparisons of S65, S66 and S69 hold on the new runs as registered.

D Theory and the Sharpening Model

Odds decomposition (Eq. 1). Let a pool contain classes $c \in \{S, A, U\}$ with shares π_c, and let $m_c(t)$ be the fraction of class c inside the band at round t. Write $I = \{A, U\}$, $\rho_S(t) = m_S(t)/m_S(0)$ and $\rho_I(t) = \sum_{c\in I} \pi_c m_c(t) / \sum_{c\in I} \pi_c m_c(0)$. The admitted ill-posed share is $s_t = \sum_I \pi_c m_c(t) / \sum_c \pi_c m_c(t)$, so $\mathrm{odds}(s_t) = \sum_I \pi_c m_c(t) / \pi_S m_S(t)$, and dividing by $\mathrm{odds}(s_0)$ gives Equation 1. □

Exit. For $K \sim \mathrm{Bin}(8, q)$, $m(q) = \Pr[K \geq 2] - \Pr[K \geq 7]$. Using $\frac{d}{dq}\Pr[K \geq k] = 8\binom{7}{k-1} q^{k-1}(1-q)^{8-k}$, $m'(q) = 56q(1-q)^6 - 56q^6(1-q) = 56q(1-q)[(1-q)^5 - q^5]$, which is negative for $q > 1/2$. Under $q_t \propto q_0^{\beta^t}$ with a unique base mode, the mode share rises monotonically in t and is larger for larger β, so after it passes $1/2$ the in-band mass falls, faster for larger β. For problems with a unique base mode and $\beta_S > \beta_I$, therefore, $\rho_S/\rho_I \to 0$, and s_t rises toward the share set by problems that never leave the band; problems whose base mode is tied never sharpen in the model and leave a floor of in-band mass in both classes. For R-Zero's filter (nine samples, counts 3–7 kept), $m'(q) = 36q^2(1-q)[7(1-q)^5 - 2q^5]$, negative for $q > 0.562$. We use the mode's count as a proxy for the largest count of the eight samples, which is what the filter reads. The two coincide at the upper edge whenever the mode reaches seven samples, so the exit argument is unaffected, but at low agreement the proxy undercounts in-band mass when competing answers repeat: for answer probabilities $(0.4, 0.3, 0.3)$ it gives 0.89 against 0.99.

Filters on top of the band. If a further filter passes a fraction ε_t of the ill-posed and k_t of the solvable problems in the band, $\mathrm{odds}(s_t^{\mathrm{ver}}) = \mathrm{odds}(s_t)\,\varepsilon_t/k_t$, so the filtered odds grow from round zero by $(\rho_I/\rho_S)\cdot(\varepsilon_t/k_t)/(\varepsilon_0/k_0)$. A filter with a constant ratio leaves the growth unchanged, and one that holds the admitted share at its round-zero level must multiply its ratio by ρ_S/ρ_I, which is 0.26 at round five on the fixed pool; a ratio that falls more slowly slows the drift without stopping it. This concerns the rounds we observe. Beyond them, since $k_t \leq 1$, a filter that keeps at least a fraction $\varepsilon > 0$ of the band's ill-posed problems admits an ill-posed share of at least $\varepsilon\rho_I/(\varepsilon\rho_I + \rho_S\pi_S m_S(0)/\sum_I \pi_c m_c(0))$, which tends to one if $\rho_S/\rho_I \to 0$. Our data do not establish this limit: ρ_S is still 0.19 at round five, and the sharpening model keeps problems with a tied base mode in the band.

Fitting. We reconstruct each problem's base answer distribution from its base agreement p_0 as $q_0 = (8p_0 + 0.5)/9$ with the remaining mass split equally over $\lceil (1-q_0)/q_0 \rceil$ competitors, apply $q_t \propto q_0^{\beta_c^t}$, and choose β_c on a grid to minimise the squared error of the in-band fraction per class over

rounds one to five. Problems whose base mode is tied never sharpen in this model (13% of solvable, 26% of ambiguous and 35% of unsolvable problems); the single-rate model uses the same reconstruction. Confidence intervals come from 1,000 bootstrap resamples within class. The held-out prediction uses the second pool's class labels and base agreements only. The reported fit is a refit in log space after an underflow in the first, linear-space script, which gave $\beta_S = 3.30$ and a held-out error of 3.0 points.

Modal count and tied modes†. The binomial in-band probability is exact if competing answers never repeat, as scattered computation errors rarely do, and undercounts in-band mass when they concentrate, as alternative readings do. Refitting with the exact probability that the largest count of a multinomial draw of eight lies in 2–6, under the most concentrated reconstruction above (the fewest competitors that leave q_0 the mode), gives $\beta_S = 7.5$ (95% CI $[3.85, 8.0]$, the upper end of the grid), $\beta_A = 1.80$ $[1.65, 2.60]$ and $\beta_U = 1.75$ $[1.65, 1.93]$; β_S exceeds both other rates in every resample, the held-out error is 2.5 points, and a single rate plateaus at 70% instead of 82%. The values of the rates therefore depend on how competing answers are modelled, but their ordering does not. Without the problems whose base mode is tied, the ordering also holds ($\beta_S = 2.40$–2.55, $\beta_A = 1.55$–1.60, $\beta_U = 1.50$–1.55 under the two in-band probabilities, in every resample), and per-class rates fit the in-band fractions of ambiguous and unsolvable problems far better than a single rate (mean absolute error 0.03–0.11 against 0.23–0.34). The model then predicts, however, that all solvable problems leave the band by round four, whereas 8–9% of them stay, so it overshoots the admitted ill-posed share after round three (to 100% against about 80%). In the full pool, the tied problems, which the model never sharpens, supply the floor that these remaining solvable problems supply in the data; the model describes which class leaves faster, not why some solvable problems are never resolved.

E Official R-Zero Replication

We ran R-Zero at commit `5699329` on Qwen3-4B-Base for three of its five default iterations with two seeds. Each iteration trains the challenger with GRPO (KL coefficient 10^{-2}; merged checkpoint at step 5). Its reward for a generated question is $\min(\hat{p}, 1-\hat{p}) - c$, where $\hat{p}$ is the majority share of ten answers from the previous solver and c is the share of the batch's questions in the question's cluster (agglomerative clustering, average linkage, distance $1 - \mathrm{BLEU}$, threshold 0.5); an output without a parsable question scores $-1 - c$. The challenger then generates 8,000 questions; each is pseudo-labelled with nine samples from the previous solver, questions whose majority share over non-empty answers lies in $[0.3, 0.8]$ are kept, and the solver is trained with GRPO (512 prompts $\times$ 5 rollouts; reward $0.9\times$ agreement with the pseudo-label $+ 0.1\times$ format; merged checkpoint at step 15, which we probe). The R-Zero paper states the filter as $|\hat{p} - \frac{1}{2}| \leq 0.25$ on ten samples (three to seven agree) and the challenger's reward as $\max(0, 1 - 2|\hat{p} - \frac{1}{2}| - r_{\mathrm{rep}})$, with 0 for a malformed output, whose uncertainty term is twice the code's; we followed the code, whose filter keeps three to seven of nine answers when all nine give one. We kept all algorithmic defaults and changed file handling (no model-hub uploads), process management (the release terminates every Python process on the host), seeding, resumption, a memory cap, and three compatibility points of newer library versions; instead of R-Zero's own benchmark evaluation we measured accuracy ourselves (Figure 8). The pinned framework versions have no kernels for our GPUs, so we used torch 2.8, vLLM 0.11 and transformers 4.57. In one seed, R-Zero's own one-hour timeout dropped 983 questions of one iteration's pseudo-labelling. We labelled 200 randomly sampled admitted questions per iteration.

Raw challenger output. For each seed and iteration we sampled 200 of the questions the challenger wrote (every generation with a question block and a boxed answer), before pseudo-labelling and filtering, shuffled the 1,200 under neutral identifiers and had them judged as in Appendix B, with a framing sentence that does not mention a filter; against the blind re-judging of the admitted questions, this framing changes the ill-posed share of the admitted questions in the sample by $+0.1$ points (CI $[-4.6, 4.8]$). Batches whose judges failed the blindness check were re-judged in up to two further passes (Appendix C, deviation 12). The raw ill-posed share is 38.5, 52.0 and 59.5% in seed 0 and 41.0, 42.5 and 48.0% in seed 1 at iterations 1–3 (changes $+21.0$ $[11.0, 30.5]$ and $+7.0$ $[-2.5, 17.0]$; pooled $+14.0$ $[7.2, 20.8]$). The admitted share of the original labels rises by 11.0 points more (CI

$[1.5, 20.5]$). Among the sampled questions that R-Zero admitted, the challenger accounts for 69% (CI $[44, 98]$) of the change in the ill-posed log-odds and downstream selection (pseudo-labelling and the filter) for the rest, a change of $+0.26$ (CI $[0.01, 0.53]$) in the log of E, the admission rate of ill-posed questions divided by that of solvable ones; E rises from 0.79 (CI $[0.63, 0.96]$) to 1.02 (CI $[0.88, 1.19]$). Computed on the pseudo-labelled questions only, which isolates the filter, the change is $+0.18$ (CI $[-0.06, 0.43]$); the filter alone admits ill-posed questions 0.87 times as often as solvable ones at the first iteration (CI $[0.70, 1.05]$). This analysis does not separate the terms of the challenger's reward.

Behaviour depends on the format. Under our plain prompt, R-Zero's final solvers answer unanswerable UMWP questions self-consistently only slightly more often than the base model (28% to 33% and 29%; paired 95% CIs of the rise $[1.3, 8.3]$ and $[-2.5, 4.0]^{\dagger}$). In the chat format they are trained in, the rise is far larger (16% to 45% and 40%†), partly because the base model often gives no answer in that format (Figure 9).

Benchmark accuracy. Figure 8 compares R-Zero's final solvers with the base model under greedy decoding. In R-Zero's chat format, the pre-registered comparison on MATH-500 shows a gain in one seed ($+4.0$ points, 95% CI $[0.8, 7.4]$) but not in the other ($+2.6$ $[-1.2, 6.2]$); under our plain prompt both seeds gain 8–10 points. Our run is a partial reproduction: in R-Zero's format, the final solvers' GSM8K accuracy matches the published value, although our base model scores lower, and their MATH-500 gain is about 0.3 of the published one, from a base model that scores 3.2 points higher.

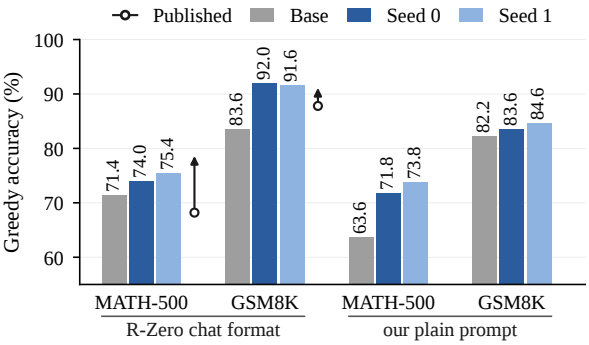


Figure 8: Greedy accuracy (%) of R-Zero's final solvers (iteration 3) and the base model. Arrows: the published base and final accuracy (Huang et al., 2026b), which use GPT-4o as answer judge. Our GSM8K set is 500 test problems.

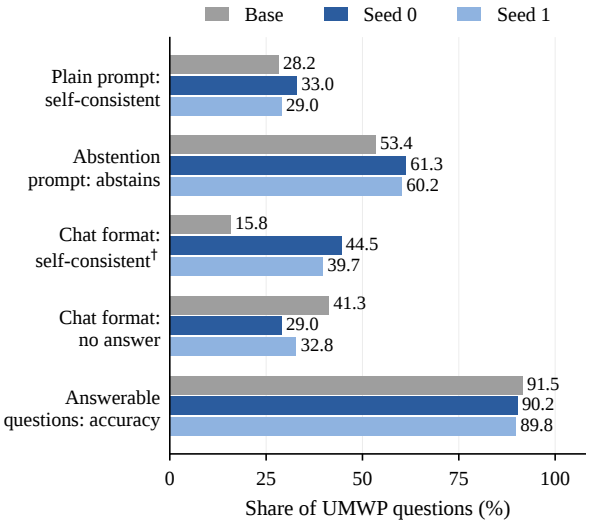


Figure 9: R-Zero's final solvers on UMWP. Under the plain prompt, the paired 95% CIs of the rise in self-consistent answers are $[1.3, 8.3]$ (seed 0) and $[-2.5, 4.0]$ (seed 1)†, from 10,000 item-bootstrap resamples.

F Further Results

Figure 10 shows the ill-posed share of every labelled round, and Figure 11 the size of every training set. *Post-trained scorers.* Post-trained models act as sharpened solvers: the band admits 75% and 76% ill-posed problems from the fixed pool when Qwen3-14B and Qwen2.5-32B-Instruct score it, against 64% and 62% for their base models (preregistered). *Reward variants.* Offering unknown without reward lowers abstention on unanswerable UMWP questions to 31–38% (full curriculum: 69–71%; base model: 53%) and raises the frontier of §6 to 23%. Letting unknown win the vote raises abstention on premise-deleted problems by 8–13 points and, in a continued loop, cuts the ill-posed problems trained toward an invented answer to 49–61% in round four (full curriculum: 79–81%), rising back to 63–75% in round five; one of three seeds then abstains on 95% of intact problems. With a triple weight, the share of training samples answering unknown reaches 100% within 22 and 40 steps, and one seed's accuracy falls from 82% to 21%.

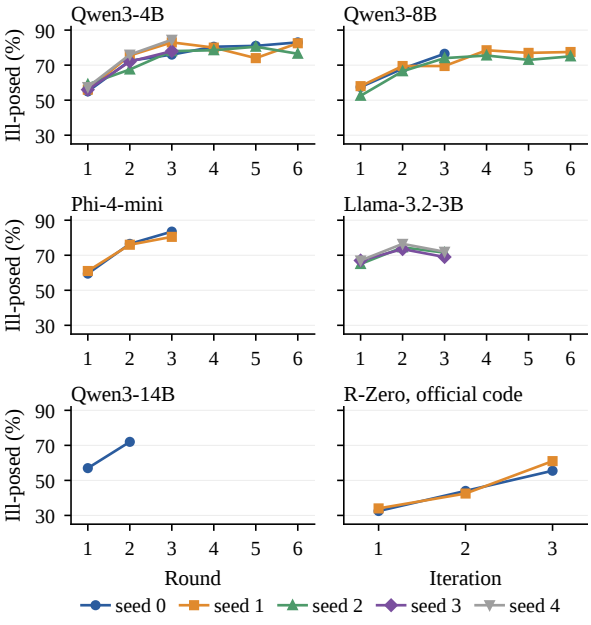


Figure 10: Ill-posed share (%) of the admitted curriculum per lineage (200 labelled problems per seed and round or iteration). Llama seeds 0 and 1 collapsed in round 1 and seed 2 after round 3 (accuracy below half the base model's, or most MATH samples truncated).

Closed loops and offline rules. The rules of Table 3 were applied offline; its classifier is a 5-fold cross-validated logistic regression on agreement under both prompts, abstention share, agreement

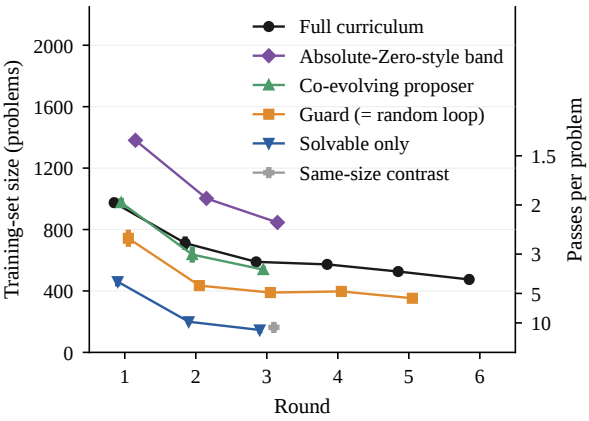


Figure 11: Training-set size of the Qwen3-4B curricula by round (bars: range over seeds; right axis: passes per problem, $1{,}920/n$). Seeds: full curriculum 0–4 (0–2 from round 4), Absolute-Zero-style band 0–1, co-evolving proposer 0–3, abstention guard 0–2, solvable only 0–3, same-size contrast 0–4 (both arms). The size-matched random loop matches the guard's size in every seed and round. Training sets include the admitted planted problems.

Inconsistency (contradictory condition). "[. . .] The pizza is cut into 8 equal slices. Emily and her 3 siblings each eat 2 slices, but 1 slice goes uneaten. If their friend Joe eats all the uneaten slices, how many more slices than Emily did he eat?"

Underspecification (missing premise). "[. . .] a square-shaped venue measures 100 feet on each side. If the venue charges $15 for each ticket and sells 70% of its total seating capacity, how much money does the venue earn from ticket sales?"

Ambiguity (ambiguous reference). "[. . .] 25 dimes in her piggy bank. She found 15 more dimes that afternoon, exchanged them for quarters, and then divided the quarters equally among her 5 grandchildren. How many cents did each grandchild receive?" (*them*: 15 or 40 dimes?)

Table 6: Problems admitted in round 3 of the Qwen3-4B loop; the kind follows the judges' notes. In a blind typing of 540 labelled ill-posed problems, contradictory or impossible conditions lead at every round (38–43%), ahead of missing premises (18–29%) and ambiguous references (14–17%).

between the prompts' answers, length and number count. As closed loops on Qwen3-4B we ran the band, the abstention guard and the size-matched random loop (seeds 0–2), the co-evolving proposer (seeds 0–3), Absolute Zero's rule (seeds 0 and 1) and a recalibrated band (seeds 0 and 1, and Llama-3.2-3B seed 1; three rounds). In each round, the recalibrated band chooses the agreement limits on the grid $k/8$, with the upper limit at most $7/8$, that maximise the admission rate of the planted intact problems minus that of their deleted twins, admitting at least 30% of the intact ones; it uses no answer labels, but the planted problems are of known status. Its admitted proposals are 29.5% and 37.0% unsolvable at round one and 50.5% and 35.0% at round three, against 64.5% and 66.5% under the fixed band, and the in-band rate of deleted relative to intact planted problems rises from 0.59 and 0.46 to 1.25 and 1.16 (Llama: 0.68 to 1.00, with 31.0% to 55.0% unsolvable). The recalibration delays and lowers the drift, but by round three the band again admits deleted problems more readily than intact ones.

Cross-seed consensus†**.** On the 205 UMWP questions that all three round-5 seeds answer self-consistently (answers read as in §2), the seeds give the same answer on 181 (88.3%, Wilson 95% interval 83.2–92.0). On 141 of these (77.9%) the common answer is the base model's most frequent answer, and on 155 (85.6%) it is one of its most frequent answers when these are tied; on 170 (93.9%) it is among the base model's eight answers, and over all 181 the base model gives it in 3.7 of eight samples on average. The match is 96 of 100 (96.0%) where the base model was itself self-consistent and 45 of 81 (55.6%) where it was not. Each seed's modal answer equals the base model's on 73–76% of the 205 questions, and the common answer equals the answer of the answerable original on 54.7%. On the 112 readings questions (fresh samples), all three seeds answer 31 self-consistently, agree on 23, and give the base model's most frequent answer on 18 of these (78.3%; 20, or 87.0%, counting ties). Probe samples that share a sampling index agree with the base model's samples at most slightly more often than samples with different indices (0.30–0.31 against 0.26–0.30).

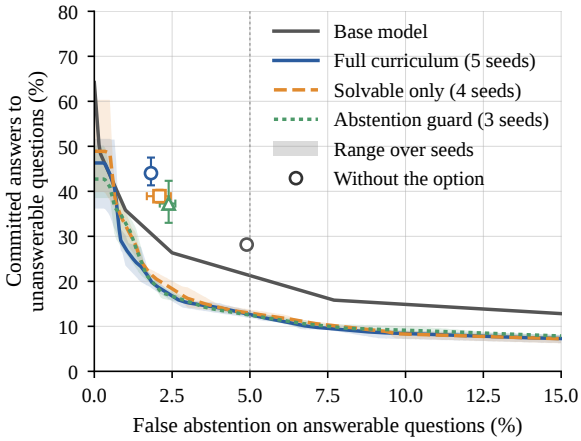


Figure 12: Selective answering on UMWP with the abstention option (Qwen3-4B, round 3). Each line is the frontier of label-free rules that answer only when enough of the eight samples agree or few of them abstain, averaged over seeds (shaded: range over seeds). Hollow markers: self-consistent answers under the plain prompt, placed at the model's false-abstention rate with the option (mean over seeds; bars: range). Dotted line: the 5% rate used in §6. Calibrated on a random half of the question pairs (stratified by source; both directions averaged)†, the rules reach 16.3–18.9% at the guard's false-abstention rate (achieved 2.2–3.4%) and 12.4–13.1% at 5% (achieved 4.4–5.6%), against 16.8–17.8% and 12.1–13.5% in sample; the base model reaches 20.7% (21.3% in sample). Over 200 random splits, the means are 16.9–18.0% and 12.2–13.0%. Under the plain prompt, the guard-trained solver answers 33–42% of the unanswerable questions self-consistently. Answers are read as in §2; the frontiers change by at most 0.2 points from the first parser's.

G Readings Annotation

We selected 112 unanswerable UMWP questions and 112 answerable ones that the base model solves in at most three of eight samples, matched on base agreement (40, 69 and 3 questions at agreement 2, 3 and 4 of 8). The base model and the round-5 solvers of three seeds each produced eight fresh solutions per question (896 units of eight solutions). LLM labellers (Claude Opus, one fresh context per batch; they saw the question and the last 1,800 characters of each solution, but not the model, round, seed or answerability) grouped the solutions of each unit into readings of the text, flagged each solution that notes missing, ambiguous or contradictory information or states an assumption it had to make, and labelled the source of disagreement (different readings, computation errors, both, or none). A second pass by fresh labellers on a random quarter of the base units agrees at κ=0.91 on the source of disagreement and κ=0.93 on the flags, and matches the number of readings exactly in 86% of units.

By category†. From base to round five (pooled over seeds), the number of readings per question and the share of flagged samples change as follows: missing key information (36 questions), 4.17 to 2.84 and 12.8% to 60.3%; ambiguous key information (58), 3.98 to 2.92 and 19.2% to 69.0%; unrealistic conditions (10), 2.80 to 2.97 and 7.5% to 38.8%; unrelated object (6), 2.83 to 2.22 and 4.2% to 31.2%; missing question (2), 3.50 to 4.00 and 1 of 16 to 6 of 48 samples.

Stricter thresholds†. Figure 13 shows the share of unanswerable UMWP questions answered self-consistently at stricter thresholds. The rise holds at every threshold and is proportionally larger at the stricter ones; ties at the top concern at most 2.2% of questions.

Exact match. The labellers' label for no disagreement ("all final answers equal") is lenient: it covers 13–37% of the unanswerable questions after five rounds, whereas all eight answers coincide on 1.8–4.5%† (none at base under either count); in the 69 further units it covers, the most frequent numeric answer holds zero to seven of the eight samples (median two), and in 50 of them at least five samples decline: the samples agree on declining. Section 4.4 reports the exact count.

Human check (pre-registered). Two external annotators labelled, independently, without AI assistance and blind to model, round and the LLM labels, in two differently ordered versions, the 184 base units not used in a first packet (92 unanswerable, 92 hard answerable; source of disagreement) and 600 single solutions (two base and two round-5 solutions for each of the 112 unanswerable and 38 answerable questions; flag and answer status). They label 88.0% of the unanswerable units as disagreeing over readings, against 16.3% of the hard answerable ones (difference +0.72, 95% CI [0.63, 0.80]; LLM labels on the same units: +0.67), and their flag rate on unanswerable questions rises from 14.3% at base to 58.0% at round five (+0.44, CI [0.36, 0.51]; LLM: +0.43): both pre-registered criteria are met. Agreement with the LLM labels is κ=0.84 and 0.76 for the two annotators on readings (binary), 0.72 and 0.59 on the four-way source and 0.96 and 0.91 on flags; between the annotators it is 0.80, 0.69 and 0.94. The annotators also judge that 29.7% of the round-5 solutions to unanswerable questions give no final numeric answer, mostly because they decline in words ("cannot be determined", "Answer: N/A"), against 2.2% under the parser our probes first used, which then read the last number of the text†. The strict reading of §2, which treats such declines, answers in a variable and solutions without a final number as non-answers, ignores text after a new question that the model writes itself, and otherwise takes the same number, agrees with the annotators on 94.6% of the solutions (first parser: 85.5%): it reads 5% of the solutions they call numeric answers as non-answers and misses 10% of their non-answers (first

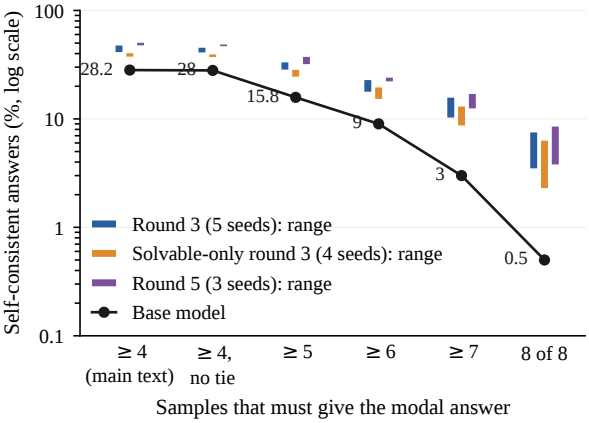


Figure 13: Self-consistent answers (%, log scale) to the 600 unanswerable UMWP questions under the plain prompt, by the number of the eight samples that must give the most frequent numeric answer (Qwen3-4B; bars: range over seeds).

parser: 2% and 92%), so it errs, if anything, toward fewer self-consistent answers. A first packet for this check was replaced by these new items before any of its labels was analysed (Appendix C, deviation 15).

Re-measurement with saved texts†. The first UMWP probes kept only the parsed answers. We therefore re-ran the same probes (prompt, sampling and seed) with texts saved on every checkpoint behind a UMWP or external-set number in the paper, 89 runs in all, and read each run with both parsers; the plan and the read scripts were logged before any new output was read, the main read script reproduces the earlier numbers exactly on the original runs, and two corrections were logged before the main read was run: the strict reading keeps the first parser's number wherever it finds one, and it cuts a solution at a new question the model writes itself, which base models often do (Qwen3-8B-Base in 53% of its samples); the second came after an interim look at completed runs, in which the uncut reading lowered Qwen3-8B-Base's accuracy on the answerable originals from 74% to 50%. Under the first parser the new runs reproduce the old ones (base 28.5% against 28.7%; round three 57–75% against 59–76%), so the differences come from the parser, not from sampling, which is not bitwise reproducible on GPUs. Under the strict reading, the base model declines in words in 3% of its samples on unanswerable questions and the round-3 solvers in 13–22%, and the self-consistent-answer rate is 28% at base, 41–48% at round three and 48–50% at round five; Qwen3-8B-Base, whose solutions the first parser often read from a question it had written itself, rises from 26% instead of 7%. All UMWP and external-set numbers in the paper come from these runs under the strict reading; registered outcomes keep the parser they were scored with (Appendix C, deviation 16). On the 112 questions of Figure 4, the strict reading changes the rate from 12.5% to 14.3% at base, from 61–75% to 40–46% at round three and from 63–79% to 41–49% at round five. Re-run so that generation itself stops at a new question, as in the loop's filter†, seven checkpoints give 30.0% for Qwen3-4B-Base (cut reading: 28.2%), 40.8% and 43.7% for two of its round-3 solvers (41.3% and 43.3%), and 28.2% and 36–41% for Qwen3-8B-Base and its round-3 solvers (26.0% and 34–38%): cutting slightly understates the base models, whose cut solutions sometimes ran on to the length limit, and the rise stays. One known flaw of the frozen reading: its decline pattern for "N/A" also matches words ending in "na", such as the name Tina, in solutions without an "Answer:" line; restricting it to a stand-alone N/A changes the self-consistent-answer rates by at most 0.2 points for Qwen3-4B and Phi-4-mini and by 0.5–0.7 points for Qwen3-8B†. The analyses of band membership (§§3–4) read answers as the loop's filter does, which stops at a new question and takes a verbal decline's last number, as the first parser does; under the strict reading their conclusions hold, but about half of the unanswerable questions that leave the band do so by declining. The secondary premise-deleted probe keeps the first parser.

H Further Related Work

Label-free self-play and rewards. R-Zero, Absolute Zero, SeRL and self-questioning models admit problems by the solver's agreement or learnability (Huang et al., 2026b; Zhao et al., 2025; Fang et al., 2025; Chen et al., 2025), TTRL rewards the majority answer (Zuo et al., 2025), grounded variants add labels or teachers (Yu et al., 2025b; Sundaram et al., 2026), and self-play extends to other tasks (Yang et al., 2026; Huang et al., 2026a; Yu et al., 2026; Cowsik et al., 2026), whose players can game their rewards (Bailey et al., 2026; Zhou, 2026). Vocabulary Dropout notes that disagreement cannot tell hard from ill-posed problems (Dineen et al., 2026); for a fixed post-trained model, DéjàQ finds high-learnability problem–answer pairs more often incorrect (Röpke et al., 2026); OpenSIR, that lowering its solve-rate threshold lowers validity, measured as agreement of GPT-5's majority answer with the reference, while validity judged by humans rises during training at its default threshold (Kwan et al., 2025); and data gating decides whether code self-play stays stable (Pu et al., 2026). Beyond these, we audit admitted sets over rounds, locate the drift in differential exit on a fixed pool, separate challenger from selection in the official R-Zero code and trace the effect on the solver's answers; we ask whether a problem's text determines a unique answer, not whether a proposed answer is correct. Other label-free objectives reward self-consistency, confidence or low entropy (Huang et al., 2023; Prasad et al., 2025; Zhao et al., 2026; Prabhudesai et al., 2025; Zhang et al., 2025; Agarwal et al., 2025), sharpening a model toward its initial mode even when wrong (Huang et al.,

2025; He et al., 2026; Shafayat et al., 2025); with labels, DAPO drops groups the solver always or never solves (Yu et al., 2025a), and spurious rewards raise Qwen accuracy (Shao et al., 2026).

Uncertainty-based selection. Sampling disagreement (Wang et al., 2023; Kuhn et al., 2023; Farquhar et al., 2024) mixes epistemic uncertainty, which training reduces, with aleatoric uncertainty, which it cannot (Kendall and Gal, 2017; Hüllermeier and Waegeman, 2021). Selection by uncertainty or loss is drawn to noisy, unlearnable points (Karamcheti et al., 2021; Mindermann et al., 2022), hence BALD's epistemic score (Houlsby et al., 2011), as curiosity is drawn to a noisy TV (Burda et al., 2019); training dynamics separate ambiguous from hard examples (Swayamdipta et al., 2020), and curricula target the frontier of learnability (Bengio et al., 2009; Jiang et al., 2021; Dennis et al., 2020). Self-play compounds this pull round after round.

Abstention, presupposition and ambiguity. QA has long studied unanswerable questions (Rajpurkar et al., 2018; Kamath et al., 2020); language models partly know what they do not know (Kadavath et al., 2022; Yin et al., 2023) and can be tuned to say so (Zhang et al., 2024; Wen et al., 2025). Reinforcement fine-tuning reduces refusal on unanswerable math (Song et al., 2025), reasoning models flag uncertainty yet still answer (Kirichenko et al., 2025; Gu et al., 2026), and abstention rewards can collapse into refusal (Che et al., 2026); these works start from labelled answerability (Tong et al., 2026), whereas the behaviour we study comes from a label-free curriculum. NLP has also studied questions with false or questionable presuppositions (Karttunen, 1974; Kim et al., 2021, 2023; Yu et al., 2023; Hu et al., 2023) and ambiguous or underspecified ones (Min et al., 2020; Liu et al., 2023; Cole et al., 2023; Hou et al., 2024; Li et al., 2025).